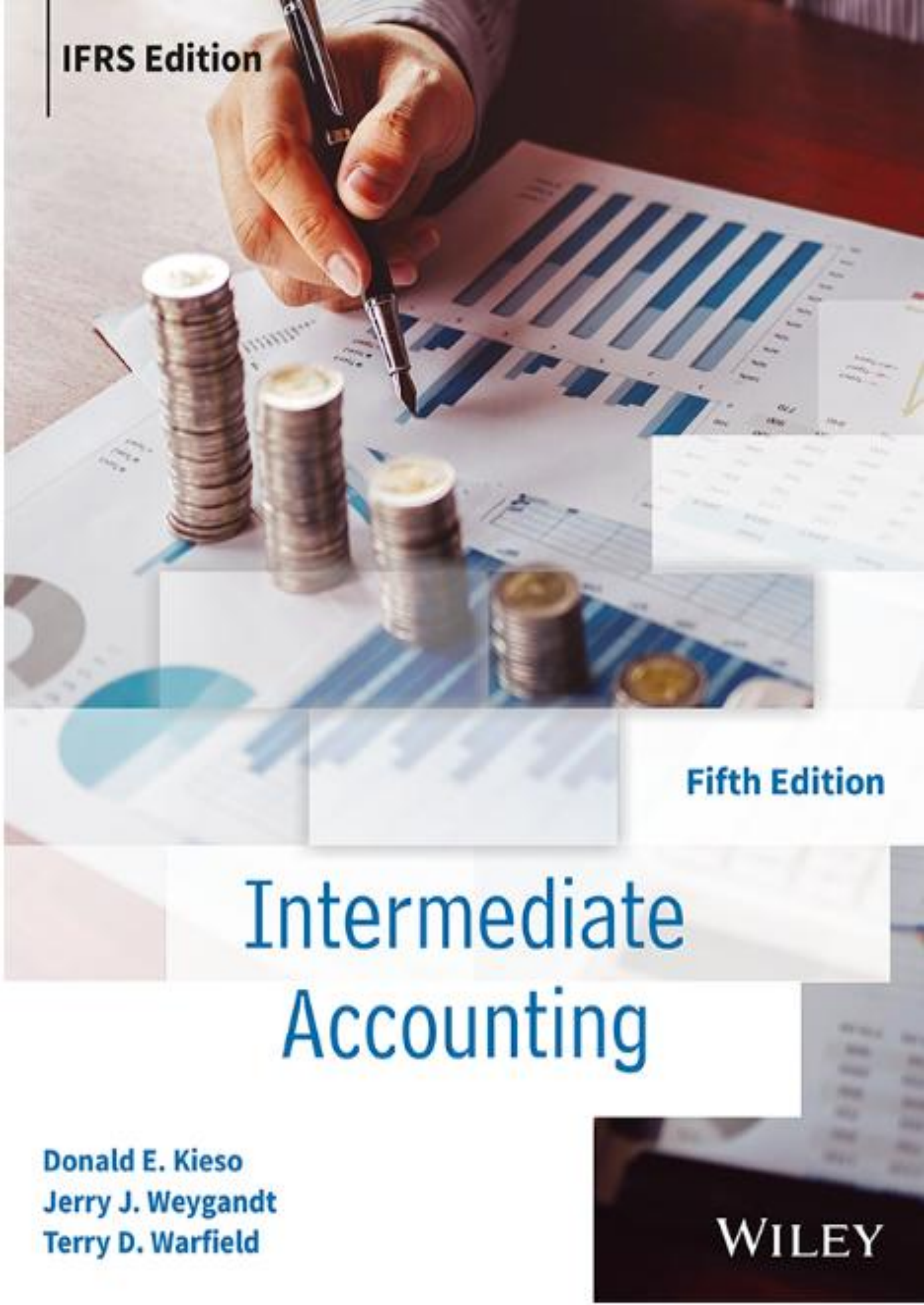


IFRS Edition

Fifth Edition

Intermediate Accounting

Donald E. Kieso
Jerry J. Weygandt
Terry D. Warfield

WILEY

CHAPTER 9

Acquisition and Disposition of Property, Plant, and Equipment

LEARNING OBJECTIVES

After studying this chapter, you should be able to:

1. Identify property, plant, and equipment and its related costs.
2. Discuss the accounting problems associated with the capitalization of borrowing costs.
3. Explain accounting issues related to acquiring and valuing plant assets.
4. Describe the accounting treatment for costs subsequent to acquisition.
5. Describe the accounting treatment for the disposal of property, plant, and equipment.

This chapter also includes numerous conceptual discussions that are integral to the topics presented here.

PREVIEW OF CHAPTER 9

As we indicate in the following opening story, market watchers monitor carefully capital expenditures on property, plant, and equipment. In this chapter, we discuss the proper accounting for the acquisition, use, and disposition of property, plant, and equipment. The content and organization of the chapter are as follows. (Note that *Global Accounting Insights* related to property, plant, and equipment is presented in Chapter 9.)

ACQUISITION AND DISPOSITION OF PROPERTY, PLANT, AND EQUIPMENT

Property, Plant, and Equipment
- Acquisition of property, plant, and equipment
- Cost of land
- Cost of land improvements
- Cost of buildings
- Cost of equipment
- Self-constructed assets

Borrowing Costs During Construction
- Qualifying assets
- Capitalization period
- Amount to capitalize
- Disclosures

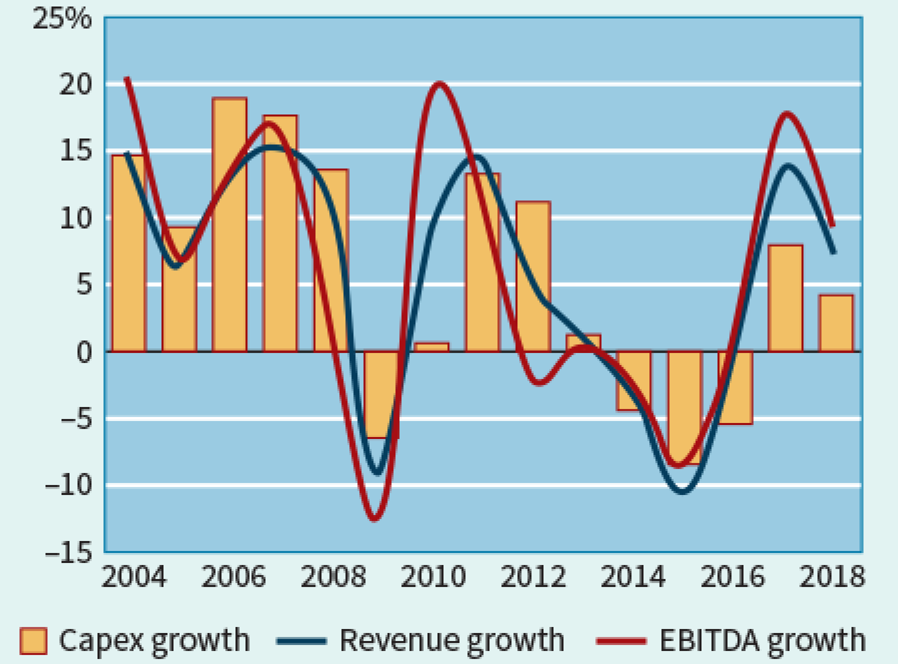
Valuation of Property, Plant, and Equipment
- Cash discounts
- Deferred-payment contracts
- Lump-sum purchases
- Issuance of shares
- Exchanges of non-monetary assets
- Government grants

Costs Subsequent to Acquisition
- Additions
- Improvements and replacements
- Rearrangement and reorganization
- Repairs
- Summary of costs subsequent to acquisition

Disposition of Property, Plant, and Equipment
- Sale of plant assets
- Involuntary conversion

Watch Your Spending

Investments in long-lived assets, such as property, plant, and equipment, are important elements in many companies' statements of financial position. Along with research and development, these investments are the driving force for many companies in generating cash flows. Property, plant, and equipment information reported on a company's statement of financial position directly affects such items as total assets, depreciation expense, cash flows, and net income. As a result, companies are careful regarding their spending level for capital expenditures (capex).



Source: S&P Global Market Intelligence.

Determining the proper level of capital expenditures in many companies is challenging. Expenditures that are too much or too little run the risk of decreasing cash flows, losing competitive position, and diminishing pricing power. The recession in Europe, the financial crisis in 2008 and its lingering effects, and China's reduced level of spending are now slowing down the growth in capital expenditures. Western Europe, Latin America, and the Asian-Pacific countries are all showing signs of capital expenditure fatigue. The adjacent graph shows that global capital expenditures are highly variable and tend to follow the trends in revenue and EBITDA growth.

Even in good times, the issues related to capital expenditures are complex. The following areas and subsequent questions have led to many sleepless nights for company managers.

1. Spending volume—how much should be spent?
2. Capital allocation—what should it be spent on?
3. Project execution—how does spending transform into real returns?

Answers to these questions are not easy, and failure to answer them correctly can lead to loss of profitability. It also follows that these issues are of extreme interest to users of financial statements. As mentioned previously, too much or too little capital spending by companies can lead to decreased cash flows, loss of competitive position, and diminished pricing power. Relationships between capital-expenditures-to-sales and capital-expenditures-to-depreciation, as well as free cash flow, provide insight into the underlying financial flexibility of a company.

Sources: *Pulling the Capex Lever*, AT Kearney (2013); S. Gordon, "Global Capital Expenditure by Companies Still Stalling," *Financial Times* (June 30, 2014); and G. Williams, "Global Corporate Capex Survey, 2019," S&P Global Ratings (June 19, 2019).

Review and Practice

Go to the **Review and Practice** section at the end of the chapter for a targeted summary review and practice problem with solution. Multiple-choice questions with annotated solutions, as well as additional exercises and practice problem with solutions, are also available online.

Property, Plant, and Equipment

LEARNING OBJECTIVE 1

Identify property, plant, and equipment and its related costs.

Companies like **Hon Hai Precision** (TWN), **Tata Steel** (IND), and **Shell plc** (GBR) use assets of a durable nature. Such assets are called **property, plant, and equipment**. Other terms commonly used are **plant assets** and **fixed assets**. We use these terms interchangeably. Property, plant, and equipment is defined as tangible assets that are held for use in production or supply of goods and services, for rentals to others, or for administrative purposes; they are expected to be used during more than one period. **[1]** (See the Authoritative Literature References section near the end of the chapter.) Property, plant, and equipment therefore includes land, building structures (offices, factories, warehouses), and equipment (machinery, furniture, tools). The major characteristics of property, plant, and equipment are as follows.

1. **They are acquired for use in operations and not for resale.** Only assets used in normal business operations are classified as property, plant, and equipment. For example, an idle building is more appropriately classified separately as an investment. Property, plant, and equipment held for possible price appreciation are classified as investments. In addition, property, plant, and equipment held for sale or disposal are separately classified and reported on the statement of financial position. Land developers or subdividers classify land as inventory.
2. **They are long-term in nature and usually depreciated.** Property, plant, and equipment yield services over a number of years. Companies allocate the

cost of the investment in these assets to future periods through periodic depreciation charges. The exception is land, which is depreciated only if a material decrease in value occurs, such as a loss in fertility of agricultural land because of poor crop rotation, drought, or soil erosion.

3. **They possess physical substance.** Property, plant, and equipment are tangible assets characterized by physical existence or substance. This differentiates them from intangible assets, such as patents or copyrights. Unlike raw material, however, property, plant, and equipment do not physically become part of a product held for resale.

Acquisition of Property, Plant, and Equipment

Most companies use historical cost as the basis for valuing property, plant, and equipment (see **Underlying Concepts**). **Historical cost** measures the cash or cash equivalent price of obtaining the asset and bringing it to the location and condition necessary for its intended use.

Underlying Concepts

Fair value is relevant to inventory but less so for property, plant, and equipment which, consistent with the going concern assumption, are held for use in the business, not for sale like inventory.

Companies recognize property, plant, and equipment when the cost of the asset can be **measured reliably** and it is probable that the company will **obtain future economic benefits**. [2] For example, when **Starbucks** (USA) purchases coffee makers for its operations, these costs are reported as assets because they can be reliably measured and benefit future periods. However, when Starbucks makes ordinary repairs to its coffee machines, it expenses these costs because the primary period benefited is only the current period.

In general, companies report the following costs as part of property, plant, and equipment. [3][1]

1. Purchase price, including import duties and non-refundable purchase taxes, less trade discounts and rebates. For example, **British Airways** (GBR) indicates that aircraft are stated at the fair value of the consideration given after offsetting manufacturing credits.
2. Costs attributable to bringing the asset to the location and condition necessary for it to be used in a manner intended by the company. For example, when **Skanska AB** (SWE) purchases heavy machinery from

Caterpillar (USA), it capitalizes the costs of purchase, including delivery costs.[2]

Companies value property, plant, and equipment in subsequent periods using either the cost method or fair value (revaluation) method. Companies can apply the cost or fair value method to all the items of property, plant, and equipment or to a single class or classes of property, plant, and equipment. For example, a company may value land (one class of asset) after acquisition using the revaluation accounting method and, at the same time, value buildings and equipment (other classes of assets) at cost.

Most companies use the cost method—it is less expensive to use because the cost of an appraiser is not needed. In addition, the revaluation (fair value) method generally leads to higher asset values, which means that companies report higher depreciation expense and lower net income. This chapter discusses the cost method; we illustrate the (fair value) revaluation method in Chapter 10.

Cost of Land

All expenditures made to acquire land and ready it for use are considered part of the land cost. Thus, when **Auchan** (FRA) or **ÆON** (JPN) purchases land on which to build a new store, land costs typically include:

1. The purchase price.
2. Closing costs, such as title to the land, attorney's fees, and recording fees.
3. Costs incurred in getting the land in condition for its intended use, such as grading, filling, draining, and clearing.
4. Assumption of any liens, mortgages, or encumbrances on the property.
5. Any additional land improvements that have an indefinite life.

For example, when ÆON purchases land for the purpose of constructing a building, it considers all costs incurred up to the excavation for the new building as land costs. **Removal of old buildings—clearing, grading, and filling—is a land cost because these activities are necessary to get the land in condition for its intended purpose**. ÆON treats any proceeds from getting the land ready for its intended use, such as salvage receipts on the demolition of an old building or the sale of cleared timber, as **reductions in the price of the land**.

In some cases, when ÆON purchases land, it may assume certain obligations on the land, such as back taxes or liens. In such situations, the cost of the land is the cash paid for it, plus the encumbrances. In other words, if the purchase price of the land is ¥50,000,000 cash but ÆON assumes accrued property taxes of ¥5,000,000 and liens of ¥10,000,000, its land cost is ¥65,000,000.

ÆON also might incur **special assessments** for local improvements, such as pavements, street lights, sewers, and drainage systems. It should charge these costs to the Land account because they are relatively permanent in nature. That is, after installation, they are maintained by the local government. In addition, ÆON should charge any permanent improvements it makes, such as landscaping, to the Land account. It records separately any **improvements with limited lives,** such as private driveways, walks, fences, and parking lots, to the **Land Improvements** account. These costs are depreciated over their estimated lives.

Generally, land is part of property, plant, and equipment. However, if the major purpose of acquiring and holding land is speculative, a company more appropriately classifies the land as an **investment**. If a real estate concern holds the land for resale, it should classify the land as **inventory**.

In cases where land is held as an investment, what accounting treatment should be given for taxes, insurance, and other direct costs incurred while holding the land? Many believe these costs should be capitalized. The reason: They are not generating revenue from the investment at this time. Companies generally use this approach except when the asset is currently producing revenue (such as rental property).

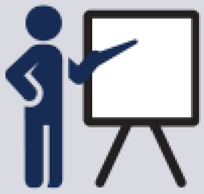
Cost of Land Improvements

Land improvements are structural additions with limited lives that are made to land. Some examples include:

- Driveways.
- Parking lots.
- Fencing.
- Lighting.
- Landscaping with a limited life, such as plants, trees, sprinkler systems, and retaining walls for planting beds.

The cost of land improvements includes all expenditures necessary to make the improvements ready for their intended use. For example, the cost of a parking lot for a new Home Depot location includes the amount paid for paving, fencing, and lighting. The company would debit the total of these costs to Land Improvements. Land improvements have limited useful lives. Even when well-maintained, they will eventually need to be replaced. As a result, companies depreciate the cost of land improvements over their useful lives.

Example 9.1 Land Costs



FACTS Assume that **Tesla** (USA) acquires property on the outskirts of Austin, Texas, for a factory site. Tesla paid $2,000,000 for the site and agreed to assume unpaid property taxes of $42,000 as well as an unpaid mortgage loan of $210,000. The city also assessed the company $38,000 for sewers, street pavement, and water mains. Tesla will be using the site as a manufacturing facility for its cybertruck. Additional costs related to the property are as follows.

Real estate commissions paid to McConaughey Realty	$120,000
Title search and title transfer fees	4,000
Costs related to demolition of an old warehouse on the land	10,000
Salvage value related to demolition of the building	1,000
Costs of surveying and grading of property	30,000
Cost of landscaping (useful life of 20 years)	101,000

QUESTION What amount should be reported for the cost of the land and any land improvements?

SOLUTION

Land costs:

Land purchase cost	$2,000,000
Assumption of unpaid property taxes	42,000
Assumption of unpaid mortgage	210,000
Assessment for sewers, pavement, and water main	38,000
Real estate commissions	120,000
Title	4,000
Demolition costs related to old warehouse	10,000
Salvage value related to old warehouse	(1,000)
Cost of surveying and grading of property	30,000
Cost of land	$2,453,000

Land improvement costs:

Cost of landscaping (limited life)	$101,000

The journal entry to record these transactions (assuming cash paid) is as follows.

Land	2,453,000
Land Improvements	101,000
Cash	2,554,000

Cost of Buildings

The cost of buildings should include all expenditures related directly to their acquisition or construction. These costs include (1) materials, labor, and overhead costs incurred during construction, and (2) professional fees and building permits.

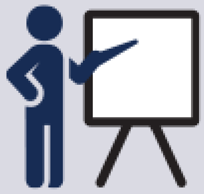
Generally, companies contract others to construct their buildings. Companies consider all costs incurred, from excavation to completion, as part of the building costs.

But how should companies account for an old building that is on the site of a newly proposed building? Is the cost of removal of the old building a cost of the land or a cost of the new building? Recall that **if a company purchases land with an old building on it, then the cost of demolition less its residual value is a cost of getting the land ready for its intended use and relates to the land**

rather than to the new building. In other words, all costs of getting an asset ready for its intended use are costs of that asset.

It follows that any costs that are not directly attributable to getting the building ready for its intended use should not be capitalized. For example, start-up costs, such as promotional costs related to the building's opening or operating losses incurred initially due to low sales, should not be capitalized. Also, general administrative expenses (such as the cost of the finance department) should not be allocated to the cost of the building.

Example 9.2 Building Costs



FACTS As a civic gesture, Art-in-Bloom is hiring an independent contractor to build a museum on land it owns. The land had been used as a park for residents in the community. To obtain a building permit for the new building, Art-in-Bloom must commit to fund the purchase of land adjacent to the museum for a new park. The CFO asks you, as the accounting intern, to determine the total cost of the building that should be capitalized on Art-in-Bloom's balance sheet based on the following information.

Payment to building contractor for cost of construction	$31,000,000
Cost of additional security during construction	80,000
Allocation of overhead cost related to Art-in-Bloom management team	200,000
Excavation costs	100,000
Cost of building permit	20,000
Funding for the cost of land for a new park	150,000
Cost of hiring a project manager to oversee the work for Art-in-Bloom	180,000
Donations to offset the cost of the building	(2,000,000)
Net total costs	$29,730,000

QUESTION At what amount should the museum be reported in the financial statements at the completion of the building?

SOLUTION

The cost of the building should include all the costs mentioned above except the allocation of overhead costs related to the management team and the donations to offset the cost of the building.

- The allocation of the overhead costs is not considered a cost of the building because these costs are not avoidable and not direct costs of the building.
- The donations should not offset the cost of the building but should be reported as revenue when received.
- The payment for funding the land for the new park should be reported as part of the cost of the building because the building could not be built if the permit were not granted.

The cost of the building is therefore $31,530,000 ($29,730,000 – $200,000 + $2,000,000) comprised of cost of construction, excavation costs, security costs, cost of building permit, cost of park land, and project manager cost.

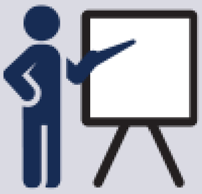
Cost of Equipment

Equipment includes assets used in operations, such as store checkout counters, office furniture, factory machinery, and delivery trucks. **Delta Air Lines'** (USA) equipment includes aircraft, in-flight entertainment systems, and trucks for ground operations. The cost of equipment consists of:

- Cash purchase price plus sales taxes.
- Freight or shipping charges.
- Insurance during transit paid by the purchaser.
- Assembling, installing, and testing the equipment.

What about recurring costs related to equipment, such as vehicle licenses and accident insurance on company trucks and cars? These items are **annual recurring expenditures and do not benefit future periods.** Therefore, these expenditures are expensed when incurred. Or, they might be recorded in a prepaid asset account if they are paid in advance.

Example 9.3 Equipment Costs-Truck



FACTS Assume that **FedEx** (USA) purchases its first battery-powered delivery truck on January 1 at a cash price of $22,000. Related expenditures are sales taxes $1,320, painting and lettering $500, vehicle license $80, and a 3-year accident insurance policy $1,600.

QUESTION What journal entry would FedEx make to record the cost of the truck?

SOLUTION

The cost of the delivery truck is $23,820, computed as follows.

Cash price	$22,000
Sales taxes	1,320
Painting and lettering	500
Cost of delivery truck	**$23,820**

FedEx treats the cost of a motor vehicle license as an expense and the cost of an insurance policy as a prepaid asset.

To record the purchase of the truck and related expenditures:

Equipment—Trucks	23,820
License Expense	80
Prepaid Insurance	1,600
Cash	25,500

Self-Constructed Assets

Occasionally, companies construct their own assets. Determining the cost of such machinery and other fixed assets can be a problem. Without a purchase price or contract price, the company must allocate costs and expenses to arrive at the cost of the **self-constructed asset**. Materials and direct labor used in construction pose no problem. A company can trace these costs directly to work and material orders related to the fixed assets constructed.

However, the assignment of indirect costs of manufacturing creates special problems. These indirect costs, called **overhead** or burden, include power, heat, light, insurance, property taxes on factory buildings and equipment, factory supervisory labor, depreciation of fixed assets, and supplies.

Companies can handle overhead in one of two ways:

1. **Assign no fixed overhead to the cost of the constructed asset.** The major argument for this treatment is that overhead is generally fixed in nature. As a result, this approach assumes that the company will have the same costs regardless of whether or not it constructs the asset. Therefore, to charge a portion of the overhead costs to the equipment will normally reduce current expenses and consequently overstate income of the current period. However, the company would assign to the cost of the constructed asset variable overhead costs that increase as a result of the construction.

2. **Assign a portion of all overhead to the construction process.** This approach, called a **full-costing approach,** assumes that costs attach to all products and assets manufactured or constructed. Under this approach, a company assigns a portion of all overhead to the construction process, as it would to normal production. Advocates say that failure to allocate overhead costs understates the initial cost of the asset and results in an inaccurate future allocation.

Companies should assign to the asset **a pro rata portion** of the fixed overhead to determine its cost. Companies use this treatment extensively because many believe that it results in a better matching of costs with revenues. Abnormal amounts of wasted material, labor, or other resources should not be added to the cost of the asset. [4]

If the allocated overhead results in recording construction costs in excess of the costs that an outside independent producer would charge, the company should record the excess overhead as a period loss rather than capitalize it. This avoids capitalizing the asset at more than its fair value. Under no circumstances should a company record a "profit on self-construction."

Borrowing Costs During Construction

LEARNING OBJECTIVE 2

Discuss the accounting problems associated with the capitalization of borrowing costs.

The proper accounting for **borrowing costs**[3] has been a long-standing controversy. Three approaches have been suggested to account for the interest incurred in financing the construction of property, plant, and equipment:

1. **Capitalize no borrowing costs during construction.** Under this approach, interest is considered a cost of financing and not a cost of construction. Some contend that if a company had used equity financing rather than debt, it would not incur this cost. The major argument against this approach is that the use of cash, whatever its source, has an associated implicit interest cost, which should not be ignored.
2. **Charge construction with all borrowing costs of funds employed, whether identifiable or not.** This method maintains that the cost of construction should include the cost of financing, whether by cash, debt, or equity. Its advocates say that all costs necessary to get an asset ready for its intended use, including borrowing costs, are part of the asset's cost. Interest, whether actual or imputed, is a cost, just as are labor and materials. A major criticism of this approach is that imputing the cost of equity capital is subjective and outside the framework of a historical cost system.
3. **Capitalize only the actual borrowing costs incurred during construction.** This approach agrees in part with the logic of the second approach—that interest is just as much a cost as are labor and materials. But, this approach capitalizes only borrowing costs incurred through debt financing. (That is, it does not try to determine the cost of equity financing.) Under this approach, a company that uses debt financing will have an asset of higher cost than a company that uses equity financing. Some consider this approach unsatisfactory because they believe the cost of an asset should be the same whether it is financed with cash, debt, or equity.

Illustration 9.1 shows how a company might add borrowing costs (if any) to the cost of the asset under the three capitalization approaches.

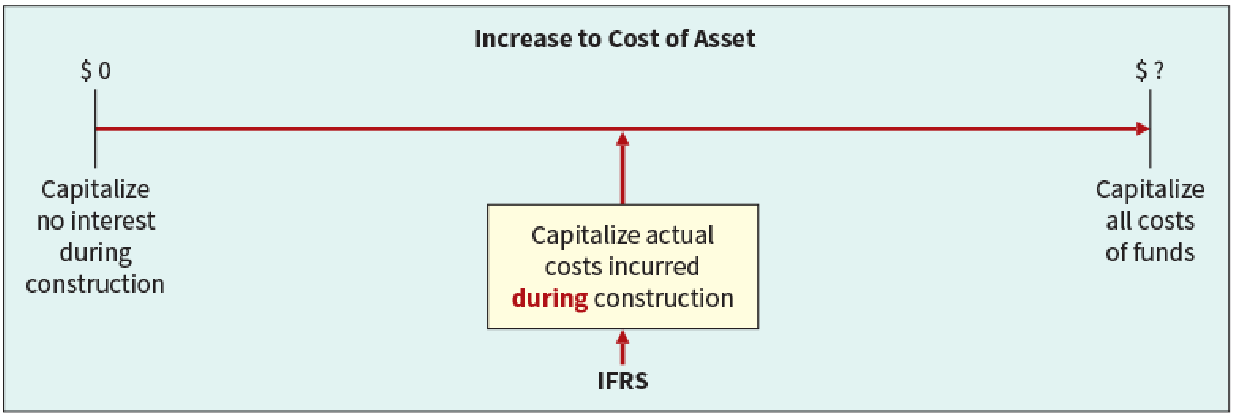


Illustration 9.1 Capitalization of Borrowing Costs

IFRS requires the third approach—capitalizing actual interest (with modifications) (see **Underlying Concepts**). This method follows the concept that the historical cost of acquiring an asset includes all costs (including borrowing costs) incurred to bring the asset to the condition and location necessary for its intended use. The rationale for this approach is that during construction, the asset is not generating revenues. Therefore, a company should defer (capitalize) borrowing costs. Once construction is complete, a company can utilize the asset in its operations. At this point, the company should report borrowing as an expense in the determination of net income. It follows that the company should expense any borrowing cost incurred in purchasing an asset that is ready for its intended use. [6]

Underlying Concepts

The objective of capitalizing borrowing costs is to obtain a measure of acquisition cost that reflects a company's total investment in the asset and to charge that cost to future periods benefitted.

To implement the capitalization approach for borrowing costs, companies consider three items:

1. Qualifying assets.
2. Capitalization period.
3. Amount to capitalize.

Qualifying Assets

To qualify for the capitalization of borrowing costs, assets must require a substantial period of time to get them ready for their intended use or sale. A company capitalizes borrowing costs starting at the beginning of the capitalization period (described in the next section). Capitalization continues until the company substantially readies the asset for its intended use.

Assets that qualify for borrowing cost capitalization include assets under construction for a company's own use (including buildings, plants, and large machinery) and assets intended for sale or lease that require a substantial period of time to produce (e.g., ships or real estate developments).[4]

Examples of assets that do not qualify for interest capitalization are (1) assets that are in use or ready for their intended use, and (2) inventories that are produced over a short period of time.

Capitalization Period

The **capitalization period** is the period of time during which a company capitalizes borrowing costs. It begins with the presence of three conditions:

1. Expenditures for the asset are being incurred.
2. Activities that are necessary to get the asset ready for its intended use or sale are in progress.
3. Borrowing cost is being incurred.

Capitalization **continues as long as these three conditions are present**. The capitalization period ends when the asset is substantially complete and ready for its intended use.

Amount to Capitalize

The amount of borrowing cost to be capitalized varies depending on whether there is **specific debt** that has been incurred to fund the project or whether the project is being funded from the **general debt** obligations of the company.

When the project is funded by specific debt, the amount capitalized is the actual borrowing costs incurred during the capitalization period offset by any investment income on temporary investments of funds made available from the borrowings.

To illustrate the issues related to the capitalization of borrowing costs funded by specific debt, assume that on November 1, 2024, Shalla Company contracted Pfeifer Construction Co. to construct a building for $1,400,000 on land costing $100,000 (purchased from the contractor and included in the first payment). Shalla made the following payments to the construction company during 2025.

January 1	March 1	May 1	December 31	Total
$210,000	$300,000	$540,000	$450,000	$1,500,000

Pfeifer Construction completed the building, ready for occupancy, on December 31, 2025. Shalla had a 15%, three-year, $1,500,000, note to finance purchase of land and construction of the building, dated December 31, 2024, with interest payable annually on December 31. During 2025, a portion of the proceeds from the borrowing that had not yet been expended in the project were invested and earned $60,000 in interest income.

The project began on January 1 and was completed on December 31, so the capitalization period was the full year of 2025. The amount of borrowing costs to be capitalized for 2025 would be computed as shown in **Illustration 9.2**.

Borrowing costs ($1,500,000 × .15)	$225,000
Investment income	60,000
Borrowing costs to be capitalized	$165,000

Illustration 9.2 Total Borrowing Costs

Shalla would make the following entries during the year.

January 1		
Land	100,000	
Buildings (or Construction in Process)	110,000	
Cash		210,000
March 1		
Buildings	300,000	
Cash		300,000
May 1		
Buildings	540,000	
Cash		540,000

December 31		
Buildings	450,000	
Cash		450,000
Buildings (Capitalized Borrowing Cost)	225,000	
Cash		225,000
Cash	60,000	
Buildings (Capitalized Borrowing Cost)		60,000

When the project is funded by general debt, some additional calculations and steps are included in the process. To illustrate, assume the same facts as the previous illustration, but, instead of any specific debt, the project is funded by the general debt of the company. Assume that Shalla had the following two debt obligations outstanding during 2025:

1. 10%, $1,000,000, five-year note payable, dated December 31, 2021, with interest payable annually on December 31.
2. 12%, $1,500,000, 10-year bonds issued December 31, 2020, with interest payable annually on December 31.

When a project is funded by general debt, the company will need to determine the **average carrying amount** of the project during the period. This can be computed as shown in **Illustration 9.3**.

Expenditures			
Date	Amount	Capitalization Period*	Average Carrying Amount
January 1	$ 210,000	12/12	$210,000
March 1	300,000	10/12	250,000
May 1	540,000	8/12	360,000
December 31	450,000	0/12	0
Totals	$1,500,000		$820,000

* The capitalization period is the number of months between when the expenditure is made and the end of the year or the end of the project, whichever occurs first.

Illustration 9.3 Average Carrying Amount Calculations

The second amount that is needed when borrowing costs from general debt are being used, and there is more than one general debt obligation, is the weightedaverage borrowing cost. This amount is called the **capitalization rate** and is computed as follows.

$$\boxed{[.10 \times (\$1,000,000 \div \$2,500,000)] + [.12 \times (\$1,500,000 \div \$2,500,000)] = 11.2\%}$$

By combining these two amounts, the amount of borrowing cost available for capitalization is now computed.

$$\boxed{\$820,000 \times .112 = \$91,840}$$

The final step when the borrowing cost of general debt is used is to apply the constraint **that the amount capitalized cannot exceed the actual borrowing costs incurred during the period.** In 2025, total borrowing costs were $280,000 [($1,000,000 × .10) + ($1,500,000 × .12)]. The amount capitalized will be lower of actual, or the amount computed by multiplying the average carrying amount by the capitalization rate. In this case, Shalla will capitalize $91,840.

All of the other entries presented above would be the same except for the interest entries on December 31, which would be as follows.

December 31		
Buildings (Capitalized Borrowing Cost)	91,840	
Interest Expense ($280,000 – $91,840)	188,160	
Cash		280,000

The final possibility is that the project is funded by a blend of specific debt and general debt. To illustrate this case, assume that Shalla has the following debt outstanding during 2025.

Specific Construction Debt

1. 15%, $750,000, three-year note to finance purchase of land and construction of the building, dated December 31, 2024, with interest payable annually on December 31. Idle funds from this borrowing were invested during early 2025 and earned $40,000 in investment income.

Other Debt

2. 10%, $500,000, five-year note payable, dated December 31, 2021, with interest payable annually on December 31.
3. 12%, $1,500,000, 10-year bonds issued December 31, 2020, with interest payable annually on December 31.

In this case, the expenditures are first allocated to the specific debt to the extent possible. Then, the remainder of the expenditures are allocated to the general debt, as shown in **Illustration 9.4**.

Date	Expenditure	Amount Allocated to Specific Borrowings	Amount Allocated to General Borrowings	Capitalization Period	Average Carrying Amount
January 1	$ 210,000	$210,000	$ 0		
March 1	300,000	300,000	0		
May 1	540,000	240,000	300,000	8/12	$200,000
December 31	450,000	0	450,000	0/12	0
Totals	$1,500,000	$750,000	$750,000		$200,000

Illustration 9.4 Allocation of Expenditures

The borrowing costs of the specific debt for this project is $72,500 based on the borrowing costs of $112,500 ($750,000 × .15) less the investment income of $40,000.

The capitalization rate on the general debt is 11.5% {[(.10 × $500,000 ÷ $2,000,000)] + [.12 × ($1,500,000 ÷ $2,000,000)]}. This results in a potential amount of borrowing costs to be capitalized of $23,000 ($200,000 × .115). Since this amount is lower than the actual borrowing costs of the general debt of $230,000 [($500,000 × .10) + ($1,500,000 × .12)], $23,000 will be capitalized from the general borrowings. The total amount to be capitalized is shown in **Illustration 9.5**.

Interest costs from specific debt	$112,500
Investment income from specific debt funds	(40,000)
Interest costs from general debt	23,000
Total borrowing costs capitalized for 2025	$ 95,500

Illustration 9.5 Summary of Borrowing Costs

At December 31, the following entry would be made.

December 31		
Buildings (Capitalized Borrowing Cost) ($112,500 + $23,000)	135,500	
Interest Expense ($230,000 − $23,000)	207,000	
Cash ($112,500 + $230,000)		342,500
Cash	40,000	
Buildings (Capitalized Borrowing Cost)		40,000

Disclosures

For each period, an entity will disclose both the amount of borrowing costs capitalized during the period and the capitalization rate used to determine the amount of borrowing costs eligible for capitalization. **Illustration 9.6** provides an example from the financial statements of **Royal Dutch Shell** (now known as **shell plc;** GBR and NLD).

Royal Dutch Shell

6 INTEREST EXPENSE

($ million)	2018	2017	2016
Interest incurred and similar charges	$3,550	$3,448	$2,732
Less: interest capitalized	(876)	(622)	(725)
Other net losses on fair value hedges of debt	169	114	4
Accretion expense	902	1,102	1,192
Total	$3,745	$4,042	$3,303

The rate applied in determining the amount of interest capitalized in 2018 was 4% (2017: 3%; 2016: 3%).

Illustration 9.6 Borrowing Costs Disclosure

Valuation of Property, Plant, and Equipment

LEARNING OBJECTIVE 3

Explain accounting issues related to acquiring and valuing plant assets.

As with other assets, **companies should record property, plant, and equipment at the fair value of what they give up or at the fair value of the asset received, whichever is more clearly evident.** However, the process of asset acquisition sometimes obscures fair value. For example, if a company buys land and buildings together for one price, how does it determine separate values for the land and buildings? We examine these types of accounting problems in the following sections.

Cash Discounts

When a company purchases plant assets subject to cash discounts for prompt payment, how should it report the discount? If it takes the discount, the company should consider the discount as a reduction in the purchase price of the asset. But should the company reduce the asset cost even if it does not take the discount?

Two points of view exist on this question. One approach considers the discount—whether taken or not—as a reduction in the cost of the asset. The rationale for this approach is that the real cost of the asset is the cash or cash equivalent price of the asset. In addition, some argue that the terms of cash discounts are so attractive that failure to take them indicates management error or inefficiency.

With respect to the second approach, its proponents argue that failure to take the discount should not always be considered a loss. The terms may be unfavorable, or it might not be prudent for the company to take the discount. At present, companies use both methods, though most prefer the former method. (*For homework purposes, treat the discount, whether taken or not, as a reduction in the cost of the asset*.)

Deferred-Payment Contracts

Companies frequently purchase plant assets on long-term credit contracts, using notes, mortgages, bonds, or equipment obligations. **To properly reflect cost, companies account for assets purchased on long-term credit contracts at the present value of the consideration exchanged between the contracting parties at the date of the transaction.**

For example, Greathouse Company purchases an asset today in exchange for a $10,000 zero-interest-bearing note payable four years from now. The company would not record the asset at $10,000. Instead, the present value of the $10,000 note establishes the exchange price of the transaction (the purchase price of the asset). Assuming an appropriate interest rate of 9% at which to discount this single payment of $10,000 due four years from now, Greathouse records this asset at $7,084.30 ($10,000 × .70843). [See Table 5.2 for the present value of a single sum, PV = $10,000 ($PVF_{4,9\%}$).]

When no interest rate is stated or if the specified rate is unreasonable, the company imputes an appropriate interest rate. The objective is to approximate the interest rate that the buyer and seller would negotiate at arm's length in a similar borrowing transaction. In imputing an interest rate, companies consider such factors as the borrower's credit rating, the amount and maturity date of the note, and prevailing interest rates. **The company uses the cash exchange price of the asset acquired (if determinable) as the basis for recording the asset and measuring the interest element.**

To illustrate, Sutter AG purchases a specially built robot spray painter for its production line. The company issues a €100,000, five-year, zero-interest-bearing note to Wrigley Robotics for the new equipment. The prevailing market rate of interest for obligations of this nature is 10%. Sutter is to pay off the note in five €20,000 installments, made at the end of each year. Sutter cannot readily determine the fair value of this specially built robot. Therefore, Sutter approximates the robot's value by establishing the fair value (present value) of the note. Entries for the date of purchase and dates of payments, plus computation of the present value of the note, are as follows.

Date of Purchase		
Equipment	75,816*	
Notes Payable		75,816

*Present value of note	= €20,000 ($PVF\text{-}OA_{5,10\%}$)
	= €20,000 (3.79079); Table 6.4
	= €75,816

End of First Year		
Interest Expense	7,582	
Notes Payable	12,418	
Cash		20,000

Interest expense in the first year under the effective-interest approach is €7,582 (€75,816 × .10). The entry at the end of the second year to record interest and principal payment is as follows.

End of Second Year		
Interest Expense	6,340	
Notes Payable	13,660	
Cash		20,000

Interest expense in the second year under the effective-interest approach is €6,340 [(€75,816 − €12,418) × .10].

If Sutter did not impute an interest rate for the deferred-payment contract, it would record the asset at an amount greater than its fair value. In addition, Sutter would understate interest expense in the income statement for all periods involved.

Lump-Sum Purchases

A common challenge in valuing fixed assets arises when a company purchases a group of assets at a single **lump-sum price**.

- When this occurs, the company allocates the total cost among the various assets on the basis of their relative fair values.
- The assumption is that costs will vary in direct proportion to fair value.
- This is the same principle that companies apply to allocate a lump-sum cost among different inventory items.

To determine fair value, a company should use valuation techniques that are appropriate in the circumstances. In some cases, a single valuation technique will be appropriate. In other cases, multiple valuation approaches might have to be used.[5]

To illustrate, Norduct Homes, Inc. decides to purchase several assets of a small heating concern, Comfort Heating, for $80,000. Comfort Heating is in the process of liquidation. Its assets sold are as follows.

	Book Value	Fair Value
Inventory	$30,000	$ 25,000
Land	20,000	25,000
Building	35,000	50,000
	$85,000	$100,000

Norduct Homes allocates the $80,000 purchase price on the basis of the relative fair values (assuming specific identification of costs is impracticable), shown in **Illustration 9.7**.

Inventory	$\frac{\$25,000}{\$100,000} \times \$80,000 = \$20,000$
Land	$\frac{\$25,000}{\$100,000} \times \$80,000 = \$20,000$
Building	$\frac{\$50,000}{\$100,000} \times \$80,000 = \$40,000$

Illustration 9.7 Allocation of Purchase Price—Relative Fair Value Basis

Issuance of Shares

When companies acquire property by issuing securities, such as ordinary shares, the par or stated value of such shares fails to properly measure the property cost. If trading of the shares is active, **the market price of the shares issued is a fair**

indication of the cost of the property acquired. The shares are a good measure of the current cash equivalent price.

For example, Upgrade Living Co. decides to purchase some adjacent land for expansion of its carpeting and cabinet operation. In lieu of paying cash for the land, the company issues to Deedland Company 5,000 ordinary shares (par value $10) that have a market price of $12 per share. Upgrade Living Co. records the following entry.

Land (5,000 × $12)	60,000	
Share Capital—Ordinary		50,000
Share Premium—Ordinary		10,000

If the company cannot determine the fair value of the ordinary shares exchanged (based on a market price), it may estimate the fair value of the property. It then uses the value of the property as the basis for recording the asset and issuance of the ordinary shares.

Exchanges of Non-Monetary Assets

The proper accounting for exchanges of non-monetary assets, such as property, plant, and equipment, is controversial.[6] Some argue that companies should account for these types of exchanges based on the fair value of the asset given up or the fair value of the asset received, with a gain or loss recognized. Others believe that they should account for exchanges based on the recorded amount (book value) of the asset given up, with no gain or loss recognized. Still others favor an approach that recognizes losses in all cases but defers gains in special situations.

Ordinarily, companies account for the exchange of **non-monetary assets** on the basis of **the fair value of the asset given up or the fair value of the asset received**, **whichever is clearly more evident**. [7] Thus, companies **should recognize immediately** any gains or losses on the exchange. The rationale for immediate recognition is that most transactions have **commercial substance**, and therefore gains and losses should be recognized.

Meaning of Commercial Substance

As indicated above, fair value is the basis for measuring an asset acquired in a non-monetary exchange if the transaction has commercial substance. An exchange has **commercial substance** if the future cash flows change as a result of the transaction. That is, if the two parties' economic positions change, the transaction has commercial substance.

For example, Andrew Co. exchanges some of its equipment for land held by Roddick Inc. It is likely that the timing and amount of the cash flows arising for the land will differ significantly from the cash flows arising from the equipment. As a result, both

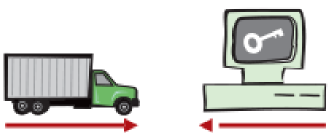
Andrew Co. and Roddick Inc. are in different economic positions. Therefore, the exchange has commercial substance, and the companies recognize a gain or loss on the exchange.

What if companies exchange similar assets, such as one truck for another truck? Even in an exchange of similar assets, a change in the economic position of the company can result. For example, let's say the useful life of the truck received is significantly longer than that of the truck given up. The cash flows for the trucks can differ significantly. As a result, the transaction has commercial substance, and the company should use fair value as a basis for measuring the asset received in the exchange.

However, it is possible to exchange similar assets but not have a significant difference in cash flows. That is, the company is in the same economic position as before the exchange. In that case, the company generally defers gains and losses on the exchange.

- Use of fair value generally results in recognizing a gain or loss at the time of the exchange.
- Consequently, companies must determine if the transaction has commercial substance.
- To make this determination, they must carefully evaluate the cash flow characteristics of the assets exchanged.[7]

Illustration 9.8 summarizes asset exchange situations and the related accounting.



Type of Exchange	Accounting Guidance
Exchange has commercial substance.	Recognize gains and losses immediately.
Exchange lacks commercial substance.	Defer gains and losses.

Illustration 9.8 Accounting for Exchanges

As Illustration 9.8 indicates, the accounting for gains and losses depends on whether the exchange has commercial substance. If the exchange has commercial substance, the company recognizes the gains and losses immediately. However, if the exchange lacks commercial substance, it defers recognition of gains and losses.

To illustrate the accounting for these different types of transactions, we examine various loss and gain exchange situations.

Exchanges—Loss Situation (Has Commercial Substance)

When a company exchanges non-monetary assets and a loss results, the company recognizes the loss if the exchange has commercial substance. The rationale:

Companies should not value assets at more than their cash equivalent price. If the loss were deferred, assets would be overstated.

For example, Information Processing SA trades its used machine for a new model at Jerrod Business Solutions NV. The exchange has commercial substance. The used machine has a book value of €8,000 (original cost €12,000 less €4,000 accumulated depreciation) and a fair value of €6,000. The new model lists for €16,000. Jerrod gives Information Processing a trade-in allowance of €9,000 for the used machine. Information Processing computes the cost of the new asset as shown in **Illustration 9.9**.

List price of new machine	€16,000
Less: Trade-in allowance for used machine	9,000
Cash payment due	7,000
Fair value of used machine	6,000
Cost of new machine	€13,000

Illustration 9.9 Computation of Cost of New Machine

Information Processing records this transaction as follows.

Equipment	13,000	
Accumulated Depreciation—Equipment	4,000	
Loss on Disposal of Equipment	2,000	
Equipment		12,000
Cash		7,000

Illustration 9.10 shows the verification of the loss on the disposal of the used machine.

Fair value of used machine	€6,000
Less: Book value of used machine	8,000
Loss on disposal of used machine	€2,000

Illustration 9.10 Computation of Loss on Disposal of Used Machine

Why did Information Processing not use the trade-in allowance or the book value of the old asset as a basis for the new equipment? The company did not use the trade-in allowance because it included a price concession (similar to a price discount). Few individuals pay list price for a new car or other new equipment. Dealers such as Jerrod often inflate trade-in allowances on the used car or equipment so that actual selling prices fall below list prices. To record the asset at list price would state it at an amount in excess of its cash equivalent price because of the new asset's inflated list price. Similarly, use of book value in this situation would overstate the value of the new machine by €2,000.[8]

Exchanges—Gain Situation (Has Commercial Substance)

Now let's consider the situation in which a non-monetary exchange has commercial substance and a gain is realized. In such a case, a company usually records the cost of a non-monetary asset acquired in exchange for another non-monetary asset at the **fair value of the asset given up**, and immediately recognizes a gain. The company should use the **fair value of the asset received** only if it is more clearly evident than the fair value of the asset given up.

To illustrate, Interstate Transportation Company exchanged a number of used trucks plus cash for a semi-truck. The used trucks have a combined book value of $42,000 (cost $64,000 less $22,000 accumulated depreciation). Interstate's purchasing agent, experienced in the secondhand market, indicates that the used trucks have a fair value of $49,000. In addition to the trucks, Interstate must pay $11,000 cash for the semi-truck. Interstate computes the cost of the semi-truck as shown in **Illustration 9.11**.

Fair value of trucks exchanged	€49,000
Cash paid	11,000
Cost of semi-truck	€60,000

Illustration 9.11 Computation of Semi-Truck Cost

Interstate records the exchange transaction as follows.

Trucks (semi)	60,000	
Accumulated Depreciation—Trucks	22,000	
Trucks (used)		64,000
Gain on Disposal of Trucks		7,000
Cash		11,000

The gain is the difference between the fair value of the used trucks and their book value. **Illustration 9.12** presents the verification of the computation.

Fair value of used trucks		$49,000
Cost of used trucks	$64,000	
Less: Accumulated depreciation	22,000	
Less: Book value of used trucks		42,000
Gain on disposal of used trucks		$ 7,000

Illustration 9.12 Computation of Gain on Disposal of Used Trucks

In this case, Interstate is in a different economic position, and therefore the transaction has commercial substance. Thus, it **recognizes a gain**.

Lacks Commercial Substance

We now assume that the Interstate Transportation Company exchange lacks commercial substance. That is, the economic position of Interstate did not change significantly as a result of this exchange. In this case, Interstate defers the gain of $7,000 and reduces the basis of the semi-truck. **Illustration 9.13** shows two different but acceptable computations to illustrate this reduction.

Fair value of semi-truck	$60,000		Book value of used trucks	$42,000
Less: Gain deferred	7,000	OR	Plus: Cash paid	11,000
Basis of semi-truck	$53,000		Basis of semi-truck	$53,000

Illustration 9.13 Basis of Semi-Truck—Fair Value vs. Book Value

Interstate records this transaction as follows.

Trucks (semi)	53,000	
Accumulated Depreciation—Trucks	22,000	
Trucks (used)		64,000
Cash		11,000

If the exchange lacks commercial substance, the company recognizes the gain (reflected in the basis of the semi-truck) when it later sells the semi-truck, not at the time of the exchange.

Illustration 9.14 presents in summary form the accounting requirements for recognizing gains and losses on exchanges of non-monetary assets.

Compute the total gain or loss on the transaction. This amount is equal to the difference between the fair value of the asset given up and the book value of the asset given up.
(a) If the exchange has commercial substance, recognize the entire gain or loss. (b) If the exchange lacks commercial substance, no gain or loss is recognized.

Illustration 9.14 Summary of Gain and Loss Recognition on Exchanges of Non-Monetary Assets

Companies disclose in their financial statements non-monetary exchanges during a period. Such disclosure indicates the nature of the transaction(s), the method of accounting for the assets exchanged, and gains or losses recognized on the exchanges.

Government Grants

Many companies receive government grants. **Government grants** are assistance received from a government in the form of transfers of resources to a company in return for past or future compliance with certain conditions relating to the operating activities of the company.[9] For example, **AB InBev NV** (BEL) received government grants related to fiscal incentives given by certain Brazilian states, based on the company's operations and investments in these states. **Danisco A/S** (DEN) notes that it receives government grants for such items as research, development, and carbon-dioxide (CO_2) allowances and investments.

In other words, a government grant is often some type of asset (such as cash; securities; property, plant, and equipment; or use of facilities) provided as a subsidy to a company. A government grant also occurs when debt is forgiven or borrowings are provided to the company at a below-market interest rate. The major accounting issues with government grants are to determine the proper method of accounting for these transfers on the company's books and how they should be presented in the financial statements.

Accounting Approaches

When companies acquire an asset such as property, plant, and equipment through a government grant, a strict cost concept dictates that the valuation of the asset should be zero. However, a departure from the historical cost principle seems justified because the only costs incurred (legal fees and other relatively minor expenditures) are not a reasonable basis of accounting for the assets acquired. To record nothing is to ignore the economic realities of an increase in wealth and assets. Therefore, most companies use the **fair value of the asset** to establish its value on the books.

What, then, is the proper accounting for the credit related to the government grant when the fair value of the asset is used?[10] Two approaches are suggested—the capital (equity) approach and the income approach. Supporters of the equity approach believe the credit should go directly to equity because often no repayment of the grant is expected. In addition, these grants are an incentive by the government—they are not earned as part of normal operations and should not offset expenses of operations on the income statement.

Supporters of the **income approach** disagree—they believe that the credit should be reported as revenue in the income statement. Government grants should not go directly to equity because the government is not a shareholder. In addition, most government grants have conditions attached to them which likely affect future expenses. They should, therefore, be reported as grant revenue (or deferred grant revenue) and matched with the associated expenses that will occur in the future as a result of the grant.

IFRS requires the income approach. The general rule is that grants should be recognized in income on a systematic basis that matches them with the related costs that they are intended to compensate. **[8]** This is accomplished in one of two ways for an asset such as property, plant, and equipment:

1. Recording the grant as deferred grant revenue, which is recognized as income on a systematic basis over the useful life of the asset, or
2. Deducting the grant from the carrying amount of the assets received from the grant, in which case the grant is recognized in income as a reduction of depreciation expense.

To illustrate application of the income approach, consider the following three examples.

Example 1: Grant for Lab Equipment

Spectrum AG received a €500,000 subsidy from the government to purchase lab equipment on January 2, 2025. The lab equipment costs €2,000,000, has a useful life of five years, and is depreciated on the straight-line basis. As indicated, Spectrum can record this grant in one of two ways: (1) Credit Deferred Grant Revenue for the subsidy and amortize the deferred grant revenue over the five-year period. (2) Credit the lab equipment for the subsidy and depreciate this amount over the five-year period.

If Spectrum chooses to record deferred revenue of €500,000, it amortizes this amount over the five-year period to income (€100,000 per year). The effects on the financial statements at December 31, 2025, are shown in **Illustration 9.15**.

Statement of Financial Position		
Non-current assets		
Equipment	€2,000,000	
Less: Accumulated depreciation	400,000	€1,600,000
Non-current liabilities		
Deferred grant revenue	€ 300,000	
Current liabilities		
Deferred grant revenue	100,000	
Income Statement		
Grant revenue for the year	€ 100,000	
Depreciation expense for the year	400,000	
Net income (loss) effect	€ (300,000)	

Illustration 9.15 Government Grant Recorded as Deferred Revenue

If Spectrum chooses to reduce the cost of the lab equipment, Spectrum reports the equipment at €1,500,000 (€2,000,000 – €500,000) and depreciates this amount over the five-year period. The effects on the financial statements at December 31, 2025, are shown in **Illustration 9.16**.

Statement of Financial Position		
Non-current assets		
Equipment	€1,500,000	
Less: Accumulated depreciation	300,000	€1,200,000
Income Statement		
Depreciation expense for the year		€ 300,000

Illustration 9.16 Government Grant Adjusted to Asset

The amount of net expense is the same for both situations (€300,000), but the presentation on the financial statements is different.[11]

Example 2: Grant for Past Losses

Flyaway Airlines has incurred substantial operating losses over the last five years. The company now has little liquidity remaining and is considering bankruptcy. The City of Plentiville does not want to lose airline service and feels it has some responsibility related to the airlines losses. It therefore agrees to provide a cash grant of $1,000,000 to the airline to pay off its creditors so that it may continue service. Because the grant is given to pay amounts owed to creditors for past losses, Flyaway Airlines should record the income in the period it is received. The entry to record this grant is as follows.

Cash	1,000,000	
Grant Revenue		1,000,000

If the conditions of the grant indicate that Flyaway must satisfy some future obligations related to this grant, then it is appropriate to credit Deferred Grant Revenue and amortize it over the appropriate periods in the future.

Example 3: Grant for Borrowing Costs

The City of Puerto Aloa is encouraging the high-tech firm TechSmart to move its plant to Puerto Aloa. The city has agreed to provide an interest-free loan of $10,000,000, with the loan payable at the end of 10 years, provided that TechSmart will employ at least 50% of its workforce from the community of Puerto Aloa over the next 10 years. TechSmart's incremental borrowing rate is 9%.

Therefore, the present value of the future loan payable ($10,000,000) is $6,499,300 ($10,000,000 × .64993$_{i=9\%,\ n=5}$). The entry to record the borrowing is as follows.

Cash	6,499,300	
Notes Payable		6,499,300

In addition, using the deferred revenue approach, the company records the grant as follows.

Cash	3,500,700	
Deferred Grant Revenue		3,500,700

TechSmart then uses the effective-interest rate to determine interest expense of $584,937 (0.09 × $6,499,300) in the first year. The company also decreases Deferred Grant Revenue and increases Grant Revenue for $584,937. As a result, the net expense related to the borrowing is zero in each year.

Unfortunately, the accounting for government grants is still somewhat unsettled. Companies are permitted to record grants at nominal values or at fair value. In addition, they may record grants to property, plant, and equipment either as a reduction of the asset or to deferred grant revenue. The key to these situations is to provide disclosures that highlight the accounting approaches. Presented below are examples of how grants are disclosed in the notes to the financial statements.

A company that adopted the deferred income approach is **PetroChina Company Limited** (CHN), as shown in **Illustration 9.17**.

PetroChina Company Limited

Note 4: Government grants

Government grants are non-reciprocal transfers of monetary or non-monetary assets from the government to the Group except for capital contributions from the government in the capacity as an investor in the Group.

A government grant is recognized when there is reasonable assurance that the grant will be received and that the Group will comply with the conditions attaching to the grant.

If a government grant is in the form of a transfer of a monetary asset, it is measured at the amount received or receivable. If a government grant is in the form of a transfer of a non-monetary asset, it is measured at fair value.

Government grants related to assets are grants whose primary condition is that the Group qualifying for them should purchase, construct, or otherwise acquire long-term assets. Government grants related to income are grants other than those related to assets. A government grant related to an asset is recognized initially as deferred income and recognized to profit or loss in a reasonable and systematic manner within the useful life of the relevant assets. A grant that compensates the Group for expenses or losses to be incurred in the future is recognized initially as deferred income, and recognized in profit or loss or released to relevant cost in the period in which the expenses or losses are recognized. A grant that compensates the Group for expenses or losses already incurred is recognized to profit or loss or released to related cost immediately.

Illustration 9.17 Deferred Income Disclosure

Daimler AG (DEU) is an example of a company adopting a policy of deducting grants related to assets from the cost of the assets, as shown in **Illustration 9.18**.

Daimler AG

Note 1: Government grants

Government grants related to assets are deducted from the carrying amount of the asset and are recognized in earnings over the life of a depreciable asset as a reduced depreciation expense. Government grants which compensate the Group for expenses are recognized as other operating income in the same period as the expenses themselves.

Illustration 9.18 Reduction of Asset Disclosure

When a company **contributes** a non-monetary asset, it should record the amount of the donation as an expense at the fair value of the donated asset. If a difference exists between the fair value of the asset and its book value, the company should recognize a gain or loss. To illustrate, Kline Industries donates land to the City of São Paulo, Brazil, for a city park. The land costs R$80,000 and has a fair value of R$110,000. Kline Industries records this donation as follows.

Contribution Expense	110,000	
Land		80,000
Gain on Disposal of Land		30,000

The gain on disposal should be reported in the "Other income and expense" section of the income statement, not as revenue.

Costs Subsequent to Acquisition

LEARNING OBJECTIVE 4

Describe the accounting treatment for costs subsequent to acquisition.

After installing plant assets and readying them for use, a company incurs additional costs over the life of the asset that range from ordinary repairs to significant additions. The major problem is allocating these costs to the proper time periods. You experience this when you own a car. **Illustration 9.19** provides examples of expenditures you make over the life of owning a car. Take a moment to study the illustration.

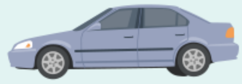
		
Examples of costs	To maintain car in normal working order: • Oil change • Repair/replace tires • Replace windshield	To increase the life or operating efficiency of the car: • New transmission • New suspension system • New battery for a hybrid car
Accounting treatment of these costs	Expense when incurred.	Capitalize (debit) the asset (car) account and depreciate over the remaining useful life of the asset.
Accounting definition	**Revenue expenditures**—Maintain a given level of service and do not increase an asset's future benefits.	**Capital expenditures**—Achieve greater future benefits for the asset, such as increased useful life, enhanced quality of output, or increased quantity of output.

Illustration 9.19 Costs over the Life of an Asset (Car)

In determining how costs should be allocated subsequent to acquisition, companies follow the same criteria used to determine the initial cost of property, plant, and equipment. That is, they recognize costs subsequent to acquisition as an asset when the costs can be measured reliably and it is probable that the company will obtain future economic benefits (see **Underlying Concepts**). Evidence of future economic benefit would include increases in (1) useful life, (2) quantity of product produced, and (3) quality of product produced.

Underlying Concepts

Expensing long-lived wastebaskets is an application of the materiality concept.

Generally, companies incur four types of major expenditures relative to existing assets.

Major Types of Expenditures

Additions. Increase or extension of existing assets.

Improvements and Replacements. Substitution of a better or similar asset for an existing one.

Rearrangement and Reorganization. Movement of assets from one location to another.

Repairs. Expenditures that maintain assets in condition for operation.

Additions

Additions should present no major accounting problems. By definition, **companies capitalize any addition to plant assets because a new asset is created**. For example, the addition of a wing to a hospital, or of an air conditioning system to an office, increases the service potential of that facility. Companies should capitalize such expenditures and record them in future periods when revenues are recognized.

One problem that arises in this area is the accounting for any changes related to the existing structure as a result of the addition. Is the cost incurred to tear down an old wall, to make room for the addition, a cost of the addition or an expense or loss of the period? The answer is that it depends on the original intent. If the company had anticipated building an addition later, then this cost of removal is a proper cost of the addition. But if the company had not anticipated this development, it should properly report the removal as a loss in the current period on the basis of inefficient planning. Conceptually, the company should remove the cost of the old wall and related depreciation and record a loss. It should then add the cost of the new wall to the cost of the building. In these situations, it is sometimes impracticable to determine a reasonable carrying amount for the old wall. Companies therefore assume the old asset to have a zero carrying amount and simply add the cost of the replacement to the overall cost.

Improvements and Replacements

Companies substitute one asset for another through **improvements** and **replacements**. What is the difference between an improvement and a replacement? An **improvement** (**betterment**) is the substitution of a **better asset** for the one currently used (say, a concrete floor for a wooden floor). A **replacement**, on the other hand, is the substitution of a **similar asset** (a wooden floor for a wooden floor).

Many times, improvements and replacements result from a general policy to modernize or rehabilitate an older building or piece of equipment. The problem is differentiating these types of expenditures from normal repairs. Does the expenditure increase the **future service potential** of the asset? Or does it merely **maintain the existing level** of service? Frequently, the answer is not clear-cut. Good judgment is required to correctly classify these expenditures.

If the expenditure increases the future service potential of the asset, a company should capitalize it. The company should simply remove the cost of the old asset and related depreciation and recognize a loss, if any. It should then add the cost of the new substituted asset.

To illustrate, Instinct Enterprises decides to replace the pipes in its plumbing system. A plumber suggests that the company use plastic tubing in place of the cast iron pipes and copper tubing. The old pipe and tubing have a book value of £15,000 (cost of £150,000 less accumulated depreciation of £135,000), and a residual value of £1,000. The plastic tubing system costs £125,000. If Instinct pays

£124,000 for the new tubing after exchanging the old tubing, it makes the following entry.

Equipment (plumbing system)	125,000	
Accumulated Depreciation—Equipment	135,000	
Loss on Disposal of Equipment	14,000	
Equipment (plumbing system)		150,000
Cash (£125,000 – £1,000)		124,000

One issue in any substitution is computing the cost and accumulated depreciation of the old asset. Fortunately, IFRS requires each significant component of an asset to be identified and its depreciation base to be determined and depreciated separately. This approach is referred to as component depreciation accounting. To illustrate, Hanoi Ltd. has a tractor that it purchased at a cost of £50,000. The following chart lists the tractor's individual components, and each component's useful life (all residual values are zero).

	Cost	Useful Life	Depreciation per Year
Tires	£ 6,000	2 years	£3,000
Transmission	10,000	5 years	2,000
Truck	34,000	10 years	3,400

Companies should keep accounting records based on the components of the asset. If the company does not have this information, estimation methods generally can provide a reasonable answer.

Rearrangement and Reorganization

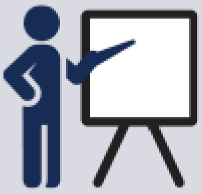
As indicated earlier, a company may incur **rearrangement** or **reorganization costs** for some of its assets. The question is whether the costs incurred in this rearrangement or reorganization are capitalized or expensed. IFRS indicates that the recognition of costs ceases once the asset is in the location and condition necessary to begin operations as management intended. As a result, the costs of reorganizing or rearranging existing property, plant, and equipment are not capitalized but are expensed as incurred.

An example is the rearrangement and reinstallation of machines to facilitate future production.

- If a company like **The Coca-Cola Company** (USA) can determine or estimate the original installation cost and the accumulated depreciation to date, it handles the rearrangement and reinstallation cost using the substitution approach.

- If not, which is generally the case, Coca-Cola should capitalize the new costs as an asset to be amortized over future periods expected to benefit.
- If these costs are immaterial, if they cannot be separated from other operating expenses, or if their future benefit is questionable, the company should immediately expense them.

Example 9.4 Rearrangement



FACTS Leonard Co. has an existing factory that it intends to demolish and redevelop. During the redevelopment period, the company will move its production facilities to another temporary site. The following costs will be incurred for the project: (1) rent of $500,000 for the temporary site, (2) removal costs of $300,000 to transport the machinery from the old location to the temporary location, and (3) $1,000,000 to install the machinery in the temporary location.

QUESTION Can all of these costs be capitalized as part of the cost of the new building? Explain.

SOLUTION

No, all the of these costs cannot be capitalized as part of the cost of the new building. Even though the costs are incremental, they are not directly attributable to the new building and it is not necessary for it to be capable of operating in the manner intended by management. The costs related to the temporary facility should be expensed as incurred.

Repairs

Ordinary Repairs

A company makes **ordinary repairs** to maintain plant assets in operating condition. It charges ordinary repairs to an expense account in the period incurred on the basis that **it is the primary period benefited**. Maintenance charges that occur regularly include replacing minor parts, lubricating and adjusting equipment, repainting, and cleaning. A company treats these as ordinary operating expenses.

It is often difficult to distinguish a repair from an improvement or replacement. The major consideration is whether the expenditure benefits more than one year or one operating cycle, whichever is longer. If a **major repair** (such as an overhaul) occurs, several periods will benefit. A company should generally handle this cost as an improvement or replacement.

Major Repairs

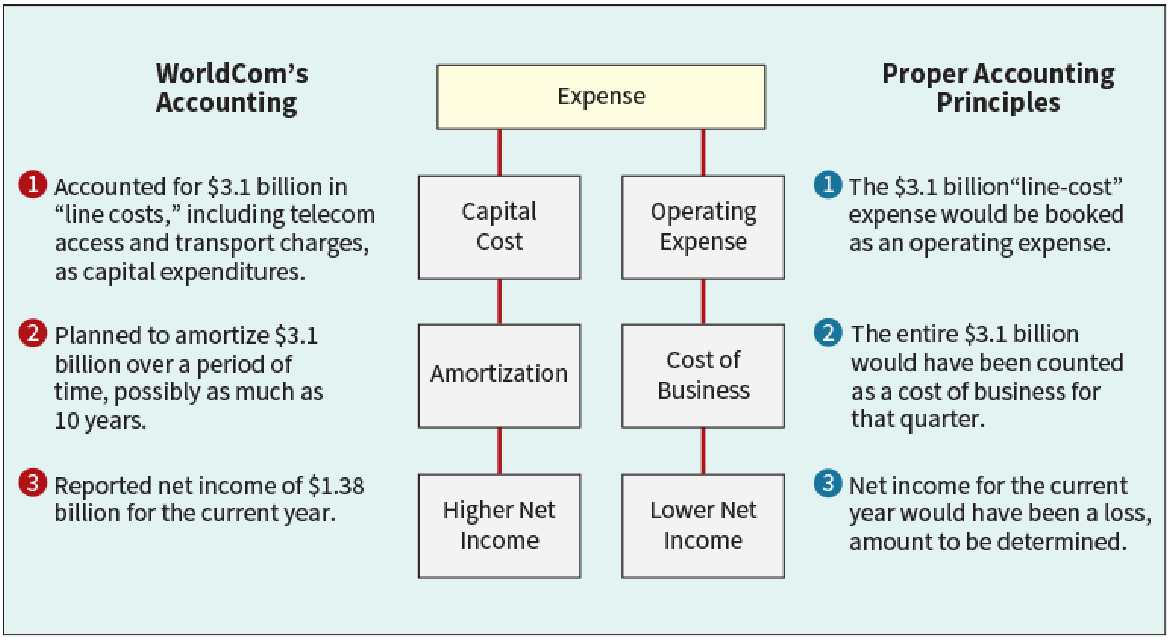
Some companies, such as airlines **Ryanair** (IRL) and **Lufthansa** (DEU), or shipping companies, such as **A.P. Møller—Maersk** (DEN) and **CMA CGM Group** (FRA), have substantial overhaul costs related to their airplanes or ships. For example, assume the Shipaway Company has just purchased a new ship for $200 million. The useful life of the ship is 20 years, but every four years it must be dry-docked and a major overhaul done. It is estimated that the overhaul will cost $4 million. The $4 million should be accounted for as a separate component of the cost of the ship (using component depreciation) and depreciated over four years. At the time of the major overhaul, the cost and related depreciation to date should be eliminated and replaced with the new cost incurred for the overhaul.

Accounting Matters

Disconnected

It all started with a check of the books by an internal auditor for **WorldCom Inc.** (USA). The telecom giant's newly installed chief executive had asked for a financial review, and the auditor was spot-checking records of capital expenditures. She found the company was using an unorthodox technique to account for one of its biggest expenses: charges paid to local telephone networks to complete long-distance calls.

Instead of recording these charges as operating expenses, WorldCom recorded a significant portion as capital expenditures. The maneuver was worth hundreds of millions of dollars to WorldCom's bottom line. It effectively turned a loss for all of one year and the first quarter of the next year into a profit. The following illustration compares WorldCom's accounting to that under proper accounting principles. Soon after this discovery, WorldCom filed for bankruptcy.



Source: Adapted from Jared Sandberg, Deborah Solomon, and Rebecca Blumenstein, "Inside WorldCom's Unearthing of a Vast Accounting Scandal," *Wall Street Journal* (June 27, 2002), p. A1; and C. Cooper, *Extraordinary Circumstances: The Journey of a Corporate Whistle Blower* (New York: John Wiley & Sons, 2008).

Summary of Costs Subsequent to Acquisition

Illustration 9.20 summarizes the accounting treatment for various costs incurred subsequent to the acquisition of capitalized assets.

Type of Expenditure	Normal Accounting Treatment
Additions	Capitalize cost of addition to asset account.
Improvements and replacements	Remove cost of and accumulated depreciation on old asset, recognizing any gain or loss. Capitalize cost of improvement/replacement.
Rearrangement and reorganization	Expense costs of rearrangement and reorganization costs as expense.
Repairs	**(a) Ordinary:** Expense cost of repairs when incurred. **(b) Major:** Remove cost and accumulated depreciation of old asset, recognizing any gain or loss. Capitalize cost of major repair.

Illustration 9.20 Summary of Costs Subsequent to Acquisition of Property, Plant, and Equipment

Disposition of Property, Plant, and Equipment

LEARNING OBJECTIVE 5

Describe the accounting treatment for the disposal of property, plant, and equipment.

A company, like **Nokia** (FIN), may retire plant assets voluntarily or dispose of them by sale, exchange, involuntary conversion, or abandonment. Regardless of the type of disposal, depreciation must be taken up to the date of disposition. Then, Nokia should remove all accounts related to the retired asset. Generally, the book value of the specific plant asset does not equal its disposal value. As a result, a gain or loss develops. The reason: Depreciation is an estimate of cost allocation and not a process of valuation. **The gain or loss is really a correction of net income** for the years during which Nokia used the fixed asset.

Nokia should report gains or losses on the disposal of plant assets in the income statement along with other items from customary business activities. However, if Nokia sold, abandoned, spun off, or otherwise disposed of the "operations of a component of a business," then it should report the results separately in the discontinued operations section of the income statement. As discussed in Chapter 3, Nokia should report any gain or loss from disposal of a business component with the related results of discontinued operations.

Sale of Plant Assets

Companies record depreciation for the period of time between the date of the last depreciation entry and the date of sale. To illustrate, assume that Barret Group recorded depreciation on a machine costing €18,000 for nine years at the rate of €1,200 per year. If it sells the machine in the middle of the 10th year for €7,000, Barret records depreciation to the date of sale as:

Depreciation Expense (€1,200 × ½)	600	
Accumulated Depreciation—Machinery		600

The entry for the sale of the asset then is:

Cash	7,000	
Accumulated Depreciation—Machinery [(€1,200 × 9) + €600]	11,400	
Machinery		18,000
Gain on Disposal of Machinery		400

The book value of the machinery at the time of the sale is €6,600 (€18,000 – €11,400). Because the machinery sold for €7,000, the amount of the gain on the sale is €400 (€7,000 – €6,600).

Involuntary Conversion

Sometimes, an asset's service is terminated through some type of **involuntary conversion**, such as fire, flood, theft, or condemnation. In some cases, companies may recover all or some of the amount lost through insurance contracts or payments from government agencies. The proper accounting for these events is accomplished in two stages. First, the loss associated with the involuntary conversion is recorded. Then, usually at a later date once the amount of the recovery has been established, the company will record a gain or revenue from the insurance proceeds or other recovery. The rationale for separating the two events is that, in most cases, both the potential and the amount of the recovery are not known at the time of the loss.[12] To record this amount at that time would result in a contingent asset, which is prohibited by IFRS. **[9]**

To illustrate, Camel Transport had a manufacturing plant located on company property that stood directly in the path of a tornado. As a result of the tornado, which occurred on May 16, 2025, the plant was completely destroyed. Camel was eventually able to reach a settlement with its insurance company for the full fair value of the plant. The settlement was reached on March 18, 2026. At the time of the tornado, the building had a carrying value of $3,500,000 (original cost of $6,000,000 less accumulated depreciation of $2,500,000). The amount of the settlement with this insurance company was $5,000,000. Camel made the following entries.

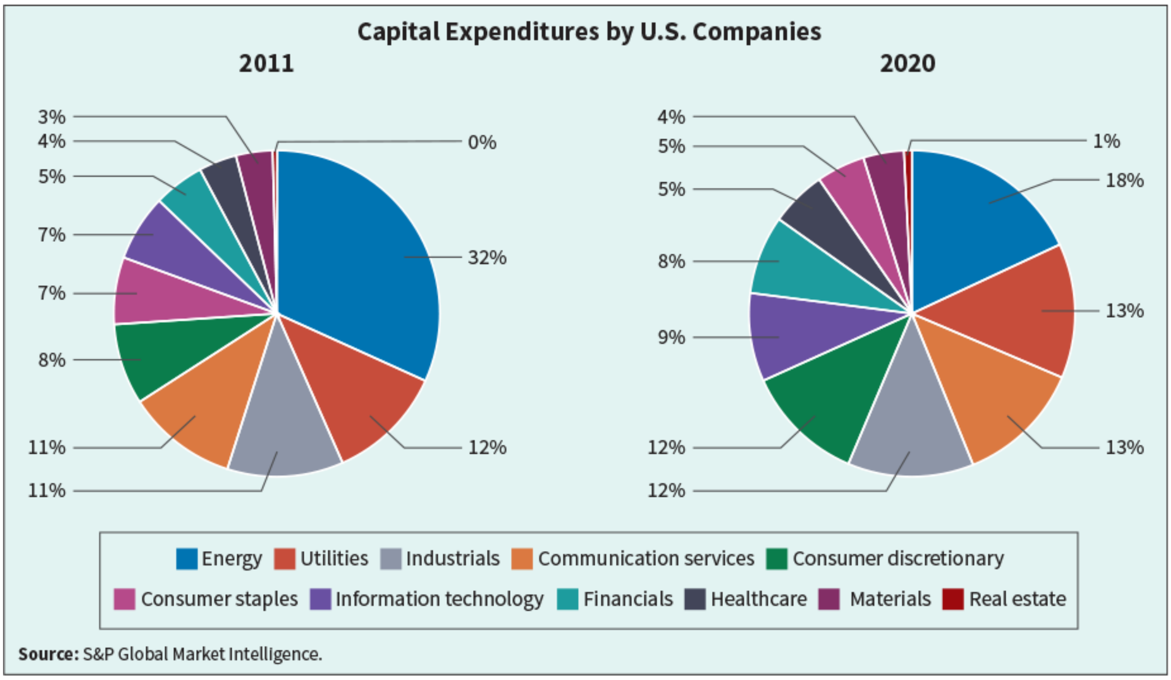
May 16, 2025		
Loss from Tornado	$3,500,000	
Accumulated Depreciation—Buildings	2,500,000	
Buildings		6,000,000
March 18, 2026		
Cash	5,000,000	
Gain from Insurance Settlement		5,000,000

Analytics in Action

Capital Expenditure Trends

Capital expenditures is an important metric for any company. It tells us the investment a company is making not only in maintaining existing assets, but whether it is investing in growth and innovation. Capital expenditures is a longer-term metric that help investors understand the strategic direction of a company. For example, Amazon.com (USA) recently invested $40 billion in capital expenditures to expand fulfillment centers and bolster technology infrastructure to support Amazon Web Services. Analyzing trends in capital expenditures over time by company and industry can provide valuable insights to financial statement users.

The following charts show a shift from 2011–2020 in which industries are making in capital expenditures.



Capital expenditures in the energy industry declined from 32% to 18%. At the same time, investments by companies in the financial, information technology, and consumer discretionary industries has increased.

Visit the Analytics in Action Activities section given in the book's product page on www.wiley.com, which includes Excel-based problems with data sets, to see the relevance and application of data analytics to intermediate accounting topics.

Review and Practice

Key Terms Review

additions 9-23

average carrying amount 9-11

borrowing costs 9-8

capitalization period 9-10

capitalization rate 9-11

commercial substance 9-16

Equipment 9-7

fixed assets 9-2

government grants 9-19

historical cost 9-3

improvements (betterments) 9-23

income approach 9-19

involuntary conversion 9-27

Land Improvements 9-4

lump-sum price 9-14

major repairs 9-25

non-monetary assets 9-15

ordinary repairs 9-25

plant assets 9-2

property, plant, and equipment 9-2

rearrangement costs 9-24

reorganization costs 9-24

replacements 9-23

self-constructed asset 9-7

Learning Objectives Review

1 Identify property, plant, and equipment and its related costs.

Property, plant, and equipment (1) are acquired for use in operations and not for resale, (2) are long-term in nature and usually subject to depreciation, and (3) possess physical substance. The costs included in **initial valuation of property, plant, and equipment** are as follows.

Cost of land: Includes all expenditures made to acquire land and to ready it for use. Land costs typically include (1) the purchase price; (2) closing costs, such as title to the land, attorney's fees, and recording fees; (3) costs incurred in getting the land in condition for its intended use, such as grading, filling, draining, and clearing; (4) assumption of any liens, mortgages, or encumbrances on the property; and (5) any additional land improvements that have an indefinite life. *Cost of land improvements:* Includes all expenditures necessary to make the improvements ready for their intended use.

Cost of buildings: Includes all expenditures related directly to their acquisition or construction. These costs include (1) materials, labor, and overhead costs incurred during construction, and (2) professional fees and building permits.

Cost of equipment: Includes the purchase price, freight and handling charges incurred, insurance on the equipment while in transit, cost of special foundations if required, assembling and installation costs, and costs of conducting trial runs.

Self-constructed assets: Indirect costs of manufacturing create special problems because companies cannot easily trace these costs directly to work and material orders related to the constructed assets. Companies might handle these costs in one of two ways: (1) assign no fixed overhead to the cost of the constructed asset, or (2) assign a portion of all overhead to the construction process. Companies should use the second method.

2 Discuss the accounting problems associated with the capitalization of borrowing costs.

Only actual borrowing costs (with modifications) should be capitalized. The rationale for this approach is that during construction, the asset is not generating revenue and therefore companies should defer (capitalize) borrowing cost. Once construction is completed, the asset is ready for its intended use and revenues can be earned. Any borrowing cost incurred in purchasing an asset that is ready for its intended use should be expensed.

3 Explain accounting issues related to acquiring and valuing plant assets.

The following issues relate to acquiring and valuing plant assets. (1) *Cash discounts:* Whether taken or not, they are generally considered a reduction in the cost of the asset; the real cost of the asset is the cash or cash equivalent price of the asset. (2) *Deferred-payment contracts:* Companies account for assets purchased on long-term credit contracts at the present value of the consideration exchanged between the contracting parties. (3) *Lump-sum purchase:* Allocate the total cost among the various assets on the basis of their relative fair values. (4) *Issuance of shares:* If the shares are actively traded, the market price of the shares issued is a fair indication of the cost of the property acquired. If the market price of the ordinary shares exchanged is not determinable, establish the fair value of the property and use it as the basis for recording the asset and issuance of the ordinary shares. (5) *Exchanges of non-monetary assets:* The accounting for exchanges of non-monetary assets depends on whether the exchange has commercial substance. See Illustrations 9.8 and 9.14 for summaries of how to account for exchanges. (6) *Grants:* Record at the fair value of the asset received, and credit deferred revenue or the appropriate asset in most cases.

4 Describe the accounting treatment for costs subsequent to acquisition.

Illustration 9.20 summarizes how to account for costs subsequent to acquisition.

5 Describe the accounting treatment for the disposal of property, plant, and equipment.

Regardless of the time of disposal, companies take depreciation up to the date of disposition and then remove all accounts related to the retired asset. Gains or losses on the retirement of plant assets are shown in the income statement along with other items that arise from customary business activities. Gains or losses on involuntary conversions are reported in "Other income and expense."

Enhanced Review and Practice

Go online for multiple-choice questions with solutions, review exercises with solutions, and a full glossary of all key terms.

Practice Problem

Columbia SA, which manufactures machine tools, had the following transactions related to plant assets in 2025.

Asset A: On June 2, 2025, Columbia purchased a stamping machine at a retail price of €12,000. Columbia paid 6% sales tax on this purchase. Columbia paid a contractor €2,800 for a specially wired platform for the machine, to ensure noninterrupted power to the machine. Columbia estimates the machine will have a 4-year useful life, with a residual value of €2,000 at the end of 4 years. The machine was put into use on July 1, 2025.

Asset B: On January 1, 2025, Columbia signed a fixed-price contract for construction of a warehouse facility at a cost of €1,000,000. It was estimated that the project will be completed by December 31, 2025. On March 31, 2025, to finance the construction cost, Columbia borrowed €1,000,000 payable April 1, 2026, plus interest at the rate of 10%. During 2025, Columbia made deposit and progress payments totaling €750,000 under the contract. The excess borrowed funds were invested in short-term securities, from which Columbia realized investment revenue of €13,000. The warehouse was completed on December 31, 2025, at which time Columbia made the final payment to the contractor. Columbia estimates the warehouse will have a 25-year useful life, with a residual value of €20,000.

Columbia uses straight-line depreciation and employs the "half-year" convention in accounting for partial-year depreciation. (Under this straight-line approach, a half-year of depreciation is recorded in the first and last year of the asset's useful life.) Columbia's fiscal year ends on December 31.

Instructions

a. At what amount should Columbia record the acquisition cost of the machine?

b. What amount of borrowing cost should Columbia include in the cost of the warehouse?

c. On July 1, 2027, Columbia decides to outsource its stamping operation to Medek Group. As part of this plan, Columbia sells the machine (and the platform) to Medek for €7,000. What is the impact of this disposal on Columbia's 2027 income before taxes?

Solution

a. Historical cost is measured by the cash or cash equivalent price of obtaining the asset and bringing it to the location and condition for its intended use. For Columbia, this is:

Price	€12,000
Tax (€12,000 × .06)	720
Platform	2,800
Total	€15,520

b.

Interest expense on specific borrowings (€1,000,000 × .10 × $^{9}/_{12}$)	€75,000
Interest revenue from excess funds	(13,000)
Borrowing cost capitalized	€62,000

c. The income effect is a gain or loss, determined by comparing the book value of the asset to the disposal value:

Cost	€15,520
Less: Accumulated depreciation (€1,690 + €3,380 + €1,690)	6,760*
Book value of machine and platform	8,760
Less: Cash received for machine and platform	7,000
Loss before income taxes	€ 1,760

*Depreciable base: €15,520 – €2,000 = €13,520. Depreciation expense: €13,520 ÷ 4 = €3,380 per year.

2025: ½year (€3,380 × .50)	€1,690
2026: full year	3,380
2027: ½year	1,690
Total	€6,760

Questions

1. What are the major characteristics of plant assets?

2. Name the items, in addition to the amount paid to the former owner or contractor, that may properly be included as part of the acquisition cost of the following plant assets.

a. Land.

b. Machinery and equipment.

c. Buildings.

3. Indicate where the following items would be shown on a statement of financial position.

a. A lien that was attached to the land when purchased.

b. Landscaping costs to ready land for use.

c. Attorney’s fees and recording fees related to purchasing land.

d. Variable overhead related to construction of machinery.

e. A parking lot servicing employees in the building.

f. Cost of temporary building for workers during construction of building.

g. Interest expense on bonds payable incurred during construction of a building.

h. Assessments for sidewalks that are maintained by the city.

i. The cost of demolishing an old building that was on the land when purchased.

4. Two positions have normally been taken with respect to the recording of fixed manufacturing overhead as an element of the cost of plant assets constructed by a company for its own use:

a. It should be excluded completely.

b. It should be included at the same rate as is charged to normal operations.

What are the circumstances or rationale that support or deny the application of these methods?

5. The Buildings account of Postera SpA includes the following items that were used in determining the basis for depreciating the cost of a building.

a. Promotion expenses.

b. Architect's fees.

c. Interest during construction.

d. Interest revenue on investments held to fund construction of a building.

e. First-year operating losses related to the building.

Do you agree with these charges? If not, how would you deal with each of these items in the company's books and in its annual financial statements?

6. Cruz Company has purchased two tracts of land. One tract will be the site of its new manufacturing plant, while the other is being purchased with the hope that it will be sold in the next year at a profit. How should these two tracts of land be reported in the statement of financial position?

7. One financial accounting issue encountered when a company constructs its own plant is whether the borrowing cost to finance construction should be capitalized and then amortized over the life of the assets constructed. What is the justification for capitalizing such interest?

8. Provide examples of assets that do not qualify for capitalization of borrowing costs.

9. What interest rates should be used in determining the amount of borrowing cost to be capitalized? How should the amount of borrowing cost to be capitalized be determined?

10. How should the amount of borrowing cost capitalized be disclosed in the notes to the financial statements? How should interest revenue from temporarily invested excess funds borrowed to finance the construction of assets be accounted for?

11. Discuss the basic accounting problem that arises in handling each of the following situations.

a. Assets purchased by issuance of ordinary shares.

b. Acquisition of plant assets by grant.

c. Purchase of a plant asset subject to a cash discount.

d. Assets purchased on a long-term credit basis.

e. A group of assets acquired for a lump sum.

f. An asset traded in or exchanged for another asset.

12. Magilke Industries acquired equipment this year to be used in its operations. The equipment was delivered by the suppliers, installed by Magilke, and placed into operation. Some of it was purchased for cash with discounts available for prompt payment. Some of it was purchased under long-term payment plans for which the interest charges approximated prevailing rates. What costs should Magilke capitalize for the new equipment purchased this year? Explain.

13. Ocampo SA purchased for €2,200,000 property that included both land and a building to be used in operations. The seller's book value was €300,000 for the land and €900,000 for the building. By appraisal, the fair value was estimated to be €500,000 for the land and €2,000,000 for the building. At what amount should Ocampo record the land and the building?

14. Pueblo Co. acquires machinery by paying $10,000 cash and signing a $5,000, 2-year, zero-interest-bearing note payable. The note has a present value of $4,208, and Pueblo purchased a similar machine last month for $13,500. At what cost should the new equipment be recorded?

15. Stan Ott is evaluating two recent transactions involving exchanges of equipment. In one case, the exchange has commercial substance. In the second situation, the exchange lacks commercial substance. Explain to Stan the differences in accounting for these two situations.

16. Crowe AG purchased a heavy-duty truck on July 1, 2022, for €30,000. It was estimated that it would have a useful life of 10 years and then would have a residual value of €6,000. The company uses the straight-line method of depreciation. It was traded on August 1, 2026, for a similar truck costing €42,000; €16,000 was allowed as trade-in value (also fair value) on the old truck and €26,000 was paid in cash. A comparison of expected cash flows for the trucks

indicates the exchange has commercial substance. What is the entry to record the exchange?

17. Ito Ltd. receives a local government grant to help defray the cost of its plant facilities. The grant is provided to encourage Ito to move its operations to a certain area. Explain how the grant might be reported.

18. Once equipment has been installed and placed in operation, subsequent expenditures relating to this equipment are frequently thought of as repairs or general maintenance and, hence, chargeable to operations in the period in which the expenditure is made. Actually, determination of whether such an expenditure should be charged to operations or capitalized involves a much more careful analysis of the character of the expenditure. What are the factors that should be considered in making such a decision?

19. What accounting treatment is normally given to the following items in accounting for plant assets?

a. Additions.

b. Major repairs.

c. Improvements.

d. Replacements.

20. New machinery, which replaced a number of employees, was installed and put in operation in the last month of the fiscal year. The employees had been dismissed after payment of an extra month's wages, and this amount was added to the cost of the machinery. Discuss the propriety of the charge. If it was improper, describe the proper treatment.

21. To what extent do you consider the following items to be proper costs of property, plant, and equipment? Give reasons for your opinions.

a. Overhead of a business that builds its own equipment.

b. Cash discounts on purchases of equipment.

c. Borrowing costs during construction of a building.

d. Profit on self-construction.

e. Freight on equipment returned before installation, for replacement by other equipment of greater capacity.

f. Cost of moving machinery to a new location.

g. Cost of plywood partitions erected as part of the remodeling of the office.

h. Replastering of a section of the building.

i. Cost of a new motor for one of the trucks.

22. Neville Enterprises has a number of fully depreciated assets that are still being used in the main operations of the business. Because the assets are fully depreciated, the president of the company decides not to show them on the statement of financial position or disclose this information in the notes. Evaluate this procedure.

23. What are the general rules for how gains or losses on disposal of plant assets should be reported in income?

Brief Exercises

BE9.1 **(LO 1)** Previn Brothers Inc. purchased land at a price of $27,000. Closing costs were $1,400. An old building was removed at a cost of $10,200. What amount should be recorded as the cost of the land?

BE9.2 **(LO 2)** Zhang Ltd. is constructing a building. Construction began on February 1 and was completed on December 31. Expenditures were HK$1,800,000 on March 1, HK$1,200,000 on June 1, and HK$3,000,000 on December 31. Compute Zhang's average carrying amount for borrowing cost capitalization purposes.

BE9.3 **(LO 2)** Zhang Ltd. (see BE9.2) had outstanding all year a 10%, 5-year HK$4,000,000 note payable and an 11%, 4-year HK$3,500,000 note payable. Compute the capitalization rate used for borrowing cost capitalization purposes.

BE9.4 **(LO 2)** Using the information from BE9.2 and BE9.3, determine the amount of borrowing cost that Zhang Ltd. would capitalize.

BE9.5 **(LO 3)** Chopra plc purchased a truck by issuing an £80,000, 4-year, zero-interest-bearing note to Equinox Inc. The market rate of interest for obligations of this nature is 10%. Prepare the journal entry to record the purchase of this truck.

BE9.6 **(LO 3)** Mohave Inc. purchased land, building, and equipment from Laguna Corporation for a cash payment of $315,000. The estimated fair values of the assets are land $60,000, building $220,000, and equipment $80,000. At what amounts should each of the three assets be recorded?

BE9.7 **(LO 3)** Fielder Company obtained land by issuing 2,000 shares of its $10 par value ordinary shares. The land was recently appraised at $85,000. The ordinary shares are actively traded at $40 per share. Prepare the journal entry to record the acquisition of the land.

BE9.8 **(LO 3)** Navajo ASA traded a used truck (cost €20,000, accumulated depreciation €18,000) for a small computer worth €3,300. Navajo also paid €500 in the transaction. Prepare the journal entry to record the exchange. (The exchange has commercial substance.)

BE9.9 **(LO 3)** Use the information for Navajo ASA from BE9.8. Prepare the journal entry to record the exchange, assuming the exchange lacks commercial substance.

BE9.10 **(LO 3)** Mehta SE traded a used welding machine (cost €9,000, accumulated depreciation €3,000) for office equipment with an estimated fair value of €5,000. Mehta also paid €3,000 cash in the transaction. Prepare the journal entry to record the exchange. (The exchange has commercial substance.)

BE9.11 **(LO 3)** Cheng SE traded a used truck for a new truck. The used truck cost $30,000 and has accumulated depreciation of $27,000. The new truck is worth $37,000. Cheng also made a cash payment of $36,000. Prepare Cheng's entry to record the exchange. (The exchange lacks commercial substance.)

BE9.12 **(LO 3)** Slaton Ltd. traded a used truck for a new truck. The used truck cost £20,000 and has accumulated depreciation of £17,000. The new truck is worth £35,000. Slaton also made a cash payment of £33,000. Prepare Slaton's entry to record the exchange. (The exchange has commercial substance.)

BE9.13 **(LO 4)** Indicate which of the following costs should be expensed when incurred.

a. €13,000 paid to rearrange and reorganize machinery.

b. €200,000 paid for addition to building.

c. €200 paid for tune-up and oil change on a delivery truck.

d. €7,000 paid to replace a wooden floor with a concrete floor.

e. €2,000 paid for a major overhaul on a truck.

BE9.14 **(LO 3)** In 2025, Sato Ltd. received a grant for ¥2 million to defray the cost of purchasing research equipment for its manufacturing facility. The total cost of the equipment is ¥10 million. Prepare the journal entry to record this transaction if Sato uses (a) the deferred revenue approach, and (b) the reduction of asset approach.

BE9.15 **(LO 5)** Ottawa Corporation owns machinery that cost $20,000 when purchased on July 1, 2019. Depreciation has been recorded at a rate of $2,400 per year, resulting in a balance in accumulated depreciation of $8,400 at December 31, 2025. The machinery is sold on September 1, 2026, for $10,500. Prepare journal entries to (a) update depreciation for 2026 and (b) record the sale.

BE9.16 **(LO 5)** Use the information presented for Ottawa Corporation in BE9.15, but assume the machinery is sold for $5,200 instead of $10,500. Prepare journal entries to (a) update depreciation for 2023 and (b) record the sale.

Exercises

E9.1 (LO 1, 2) (Acquisition Costs of Realty) The expenditures and receipts below are related to land, land improvements, and buildings acquired for use in a business. The receipts are enclosed in parentheses.

a.	Money borrowed to pay building contractor (signed a note)	€(275,000)
b.	Payment for construction from note proceeds	275,000
c.	Cost of land fill and clearing	10,000
d.	Delinquent real estate taxes on property assumed by purchaser	7,000
e.	Premium on 6-month insurance policy during construction	6,000
f.	Refund of 1-month insurance premium because construction completed early	(1,000)
g.	Architect's fee on building	25,000
h.	Cost of real estate purchased as a plant site (land €200,000 and building €50,000)	250,000
i.	Commission fee paid to real estate agency	9,000
j.	Installation of fences around property	4,000
k.	Cost of razing and removing building	11,000
l.	Proceeds from residual value of demolished building	(5,000)
m.	Borrowing costs incurred during construction on money borrowed for construction	13,000
n.	Cost of parking lots and driveways	19,000
o.	Cost of trees and shrubbery planted (permanent in nature)	14,000
p.	Excavation costs for new building	3,000

Instructions

Identify each item by letter and list the items in columnar form, using the headings shown below. All receipt amounts should be reported in parentheses. For any amounts entered in the Other Accounts column, also indicate the account title.

Item	Land	Land Improvements	Buildings	Other Accounts

E9.2 (LO 1, 2) (Acquisition Costs of Realty) Pollachek Co. purchased land as a factory site for $450,000. The process of tearing down two old buildings on the site and constructing the factory required 6 months. The company paid $42,000 to raze the old buildings and sold salvaged lumber and brick for $6,300. Legal fees of

$1,850 were paid for title investigation and drawing the purchase contract. Pollachek paid $2,200 to an engineering firm for a land survey, and $65,000 for drawing the factory plans. The land survey had to be made before definitive plans could be drawn. Title insurance on the property cost $1,500, and a liability insurance premium paid during construction was $900. The contractor's charge for construction was $2,740,000. The company paid the contractor in two installments: $1,200,000 at the end of 3 months and $1,540,000 upon completion. Borrowing costs of $170,000 were incurred to finance the construction.

Instructions

Determine the cost of the land and the cost of the building as they should be recorded on the books of Pollachek Co. Assume that the land survey was for the building.

E9.3 (LO 1) (Acquisition Costs of Trucks) Haddad Corporation operates a retail computer store. To improve delivery services to customers, the company purchases four new trucks on April 1, 2025. The terms of acquisition for each truck are described below.

1. Truck #1 has a list price of $15,000 and is acquired for a cash payment of $13,900.
2. Truck #2 has a list price of $20,000 and is acquired for a down payment of $2,000 cash and a zero-interest-bearing note with a face amount of $18,000. The note is due April 1, 2026. Haddad would normally have to pay interest at a rate of 10% for such a borrowing, and the dealership has an incremental borrowing rate of 8%.
3. Truck #3 has a list price of $16,000. It is acquired in exchange for a computer system that Haddad carries in inventory. The computer system cost $12,000 and is normally sold by Haddad for $15,200. Haddad uses a perpetual inventory system.
4. Truck #4 has a list price of $14,000. It is acquired in exchange for 1,000 ordinary shares in Haddad Corporation. The shares have a par value per share of $10 and a market price of $13 per share.

Instructions

Prepare the appropriate journal entries for the foregoing transactions for Haddad Corporation. (Round computations to the nearest dollar.)

E9.4 (LO 1, 2) (Purchase and Self-Constructed Cost of Assets) Dane ASA both purchases and constructs various equipment it uses in its operations. The following items for two different types of equipment were recorded in random order during the calendar year 2025.

Purchase	
Cash paid for equipment, including sales tax of €5,000	€105,000
Freight and insurance cost while in transit	2,000
Cost of moving equipment into place at factory	3,100
Wage cost for technicians to test equipment	6,000
Insurance premium paid during first year of operation on this equipment	1,500
Special plumbing fixtures required for new equipment	8,000
Repair cost incurred in first year of operations related to this equipment	1,300
Construction	
Material and purchased parts (gross cost €200,000; failed to take 1% cash discount)	€200,000
Imputed interest on funds used during construction (share financing)	14,000
Labor costs	190,000
Allocated overhead costs (fixed—€20,000; variable—€30,000)	50,000
Profit on self-construction	30,000
Cost of installing equipment	4,400

Instructions

Compute the total cost for each of these two types of equipment. If an item is not capitalized as a cost of the equipment, indicate how it should be reported.

E9.5 (LO 1, 2) (Treatment of Various Costs) Allegro Supply plc, a newly formed company, incurred the following expenditures related to Land, Buildings, and Equipment.

Abstract company's fee for title search		£ 520
Architect's fees		3,170
Cash paid for land and dilapidated building thereon		92,000
Removal of old building	£20,000	
Less: Residual value	5,500	14,500
Interest on short-term loans during construction		7,400
Excavation before construction for basement		19,000

Equipment purchased (subject to 2% cash discount, which was not taken)		65,000
Freight on equipment purchased		1,340
Storage charges on equipment, necessitated by non-completion of building when equipment was delivered		2,180
New building constructed (building construction took 6 months from date of purchase of land and old building)		485,000
Assessment by city for drainage project		1,600
Hauling charges for delivery of equipment from storage to new building		620
Installation of equipment		2,000
Trees, shrubs, and other landscaping after completion of building (permanent in nature)		5,400

Instructions

Determine the amounts that should be debited to Land, Buildings, and Equipment. Indicate how any costs not debited to these accounts should be recorded.

E9.6 (LO 1, 2, 3) (Correction of Improper Cost Entries) Plant acquisitions for selected companies are presented as follows.

1. Natchez Industries Inc. acquired land, buildings, and equipment from a bankrupt company, Vivace Co., for a lump-sum price of $680,000. At the time of purchase, Vivace's assets had the following book and appraisal values.

	Book Values	Appraisal Values
Land	$200,000	$150,000
Buildings	230,000	350,000
Equipment	300,000	300,000

To be conservative, the company decided to take the lower of the two values for each asset acquired. The following entry was made.

Land	150,000	
Buildings	230,000	
Equipment	300,000	
Cash		680,000

2. Arawak Enterprises purchased store equipment by making a $2,000 cash down payment and signing a 1-year, $23,000, 10% note payable. The purchase was recorded as follows.

Equipment	27,300	
Cash		2,000
Notes Payable		23,000
Interest Payable		2,300

3. Ace Company purchased office equipment for $20,000, terms 2/10, n/30. Because the company intended to take the discount, it made no entry until it paid for the acquisition. The entry was:

Equipment	20,000	
Cash		19,600
Purchase Discounts		400

4. Paunee Inc. recently received at zero cost a building from the Village of Cardassia as an inducement to locate its business in the village. The appraised value of the building is $270,000. The company made no entry to record the building because it had no cost basis.

5. Mohegan Company built a warehouse for $600,000. It could have purchased the building for $740,000. The controller made the following entry.

Buildings	740,000	
Cash		600,000
Profit on Construction		140,000

Instructions

Prepare the entry that should have been made at the date of each acquisition.

E9.7 **(LO 2)** **(Capitalization of Borrowing Costs)** McPherson Furniture started construction of a combination office and warehouse building for its own use at an estimated cost of €5,000,000 on January 1, 2025. McPherson expected to complete the building by December 31, 2025. McPherson has the following debt obligations outstanding during the construction period.

Construction loan—12% interest, payable semiannually, issued December 31, 2024	€2,000,000
Short-term loan—10% interest, payable monthly, and principal payable at maturity on May 30, 2026	1,600,000

Long-term loan—11% interest, payable on January 1 of each year. Principal payable on January 1, 2029	1,000,000

Instructions

(Carry all computations to two decimal places.)

a. Assume that McPherson completed the office and warehouse building on December 31, 2025, as planned at a total cost of €5,200,000. The following expenditures were made during the period for this project: January 1, €1,000,000; April 1, €1,500,000; July 1, €2,000,000; and October 1, €700,000. Excess funds from the construction loans were invested during the period and earned €20,000 of investment income. Compute the amount of borrowing costs to be capitalized for this project.

b. Compute the depreciation expense for the year ended December 31, 2026. McPherson elected to depreciate the building on a straight-line basis and determined that the asset has a useful life of 30 years and a residual value of €300,000.

E9.8 (LO 2) **(Capitalization of Borrowing Costs)** On December 31, 2024, Tsang Group borrowed HK$3,000,000 at 12% payable annually to finance the construction of a new building. In 2025, the company made the following expenditures related to this building: March 1, HK$360,000; June 1, HK$600,000; July 1, HK$1,500,000; and December 1, HK$1,200,000. Additional information is provided as follows.

1. Other debt outstanding

10-year, 11% bond, December 31, 2018, interest payable annually	HK$4,000,000
6-year, 10% note, dated December 31, 2022, interest payable annually	HK$1,600,000

2. March 1, 2025, expenditure included land costs of HK$150,000
3. Interest revenue earned in 2025 on funds related to specific borrowing HK$49,000

Instructions

a. Determine the amount of borrowing cost to be capitalized in 2025 in relation to the construction of the building.

b. Prepare the journal entry to record the capitalization of borrowing costs and the recognition of interest expense, if any, at December 31, 2025.

E9.9 (LO 2) (Capitalization of Borrowing Costs) On July 31, 2025, Bismarck Company engaged Duval Tooling Company to construct a special-purpose piece of factory machinery. Construction began immediately and was completed on November 1, 2025. To help finance construction, on July 31 Bismarck issued a $400,000, 3-year, 12% note payable at Wellington National Bank, on which interest is payable each July 31. On July 31, $300,000 of the proceeds of the note was paid to Duval. The remainder of the proceeds was temporarily invested in short-term marketable securities (trading securities) at 10% until November 1. On November 1, Bismarck made a final $100,000 payment to Duval.

Instructions

a. Calculate the interest expense, interest revenue, and total borrowing cost to be capitalized during 2025. (Round all computations to the nearest dollar.)

b. Prepare the journal entries needed on the books of Bismarck Company at each of the following dates.

   1. July 31, 2025.
   2. November 1, 2025.
   3. December 31, 2025.

E9.10 (LO 2) (Capitalization of Borrowing Costs) The following two situations involve the capitalization of borrowing costs.

Situation I: On January 1, 2025, Columbia Outfitters signed a fixed-price contract to have Builder Associates construct a major plant facility at a cost of R$4,000,000. It was estimated that it would take 3 years to complete the project. Also on January 1, 2025, to finance the construction cost, Columbia borrowed R$4,000,000 payable in 10 annual installments of R$400,000, plus interest at the rate of 10%. During 2025, Columbia made deposit and progress payments totaling R$1,500,000 under the contract. The excess borrowed funds were invested in short-term securities, from which Columbia realized investment income of R$50,000.

Instructions

What amount should Columbia report as capitalized borrowing cost at December 31, 2025?

Situation II: During 2025, Evander SA constructed and manufactured certain assets and incurred the following borrowing costs in connection with those activities.

	Borrowing Costs Incurred
Warehouse constructed for Evander's own use	R$30,000

Special-order machine for sale to unrelated customer, produced according to customer's specifications	9,000
Inventories routinely manufactured, produced on a repetitive basis	8,000

All of these assets required an extended period of time for completion.

Instructions

Assuming the effect of borrowing cost capitalization is material, what is the total amount of borrowing cost to be capitalized?

E9.11 (LO 3) (Entries for Equipment Acquisitions) Song Engineering purchased conveyor equipment with a list price of 15,000. Presented below are three independent cases related to the equipment (amounts in thousands).

a. Song paid cash for the equipment 8 days after the purchase. The vendor's credit terms are 2/10, n/30. Assume that equipment purchases are initially recorded gross.

b. Song traded in equipment with a book value of 2,000 (initial cost 8,000), and paid 14,200 in cash one month after the purchase. The old equipment could have been sold for 400 at the date of trade. (The exchange has commercial substance.)

c. Song gave the vendor a 16,200, zero-interest-bearing note for the equipment on the date of purchase. The note was due in 1 year and was paid on time. Assume that the effective-interest rate in the market was 9%.

Instructions

Prepare the general journal entries required to record the acquisition and payment in each of the independent cases above. (Round to the nearest won.)

E9.12 (LO 1, 3) (Entries for Asset Acquisition, Including Self-Construction) Below are transactions related to Impala Company.

a. The City of Pebble Beach gives the company 5 acres of land as a plant site. The fair value of this land is determined to be $81,000.

b. A total of 14,000 ordinary shares with a par value of $50 per share are issued in exchange for land and buildings. The property has been appraised at a fair value of $810,000, of which $180,000 has been allocated to land and $630,000 to buildings. The shares of Impala Company are not listed on any exchange, but a block of 100 shares was sold by a shareholder 12 months ago at $65 per share, and a block of 200 shares was sold by another shareholder 18 months ago at $58 per share.

c. No entry has been made to remove from the accounts for Materials, Direct Labor, and Overhead the amounts properly chargeable to plant asset accounts for machinery constructed during the year. The following information is given relative to costs of the machinery constructed.

Materials used	$12,500
Factory supplies used	900
Direct labor incurred	16,000
Additional overhead (over regular) caused by construction of machinery, excluding factory supplies used	2,700
Fixed overhead rate applied to regular manufacturing operations	60% of direct labor cost
Cost of similar machinery if it had been purchased from outside suppliers	44,000

Instructions

Prepare journal entries on the books of Impala Company to record these transactions.

E9.13 (LO 1, 3) (Entries for Acquisition of Assets) The following information is related to Rommel Group.

1. On July 6, Rommel acquired the plant assets of Studebaker SE, which had discontinued operations. The appraised value of the property is:

Land	€ 400,000
Building	1,200,000
Machinery	800,000
Total	€2,400,000

Rommel gave 12,500 shares of its €100 par value ordinary shares in exchange. The shares had a market price of €180 per share on the date of the purchase of the property.

2. Rommel expended the following amounts in cash between July 6 and December 15, the date when it first occupied the building.

Repairs to building	€105,000
Construction of bases for machinery to be installed later	135,000
Driveways and parking lots	122,000
Remodeling of office space in building, including new partitions and walls	161,000

Special assessment by city on land	18,000

3. On December 20, the company paid cash for machinery, €280,000, subject to a 2% cash discount, and freight on machinery of €10,500.

Instructions

Prepare entries on the books of Rommel for these transactions.

E9.14 (LO 3) (Purchase of Equipment with Zero-Interest-Bearing Debt) Sterling Inc. has decided to purchase equipment from Central Industries on January 2, 2025, to expand its production capacity to meet customers' demand for its product. Sterling issues a $900,000, 5-year, zero-interest-bearing note to Central for the new equipment when the prevailing market rate of interest for obligations of this nature is 12%. The company will pay off the note in five $180,000 installments due at the end of each year over the life of the note.

Instructions

a. Prepare the journal entry or entries at the date of purchase. (Round to nearest dollar in all computations.)
b. Prepare the journal entry or entries at the end of the first year to record the payment and interest, assuming that the company employs the effective-interest method.
c. Prepare the journal entry or entries at the end of the second year to record the payment and interest.
d. Assuming that the equipment had a 10-year life and no residual value, prepare the journal entry necessary to record depreciation in the first year. (Straight-line depreciation is employed.)

E9.15 (LO 3) (Purchase of Computer with Zero-Interest-Bearing Debt) Windsor plc purchased a computer on December 31, 2024, for £130,000, paying £30,000 down and agreeing to pay the balance in five equal installments of £20,000 payable each December 31 beginning in 2025. An assumed interest rate of 10% is implicit in the purchase price.

Instructions

a. Prepare the journal entry or entries at the date of purchase. (Round to two decimal places.)
b. Prepare the journal entry or entries at December 31, 2025, to record the payment and interest (effective-interest method employed).
c. Prepare the journal entry or entries at December 31, 2026, to record the payment and interest (effective-interest method employed).

E9.16 (LO 2, 3) Groupwork (Asset Acquisition) Logan Industries purchased the following assets and constructed a building as well. All this was done during the current year.

Assets 1 and 2: These assets were purchased at a lump sum for €104,000 cash. The following information was gathered.

Description	Initial Cost on Seller's Books	Depreciation to Date on Seller's Books	Book Value on Seller's Books	Appraised Value
Machinery	€100,000	€50,000	€50,000	€90,000
Equipment	60,000	10,000	50,000	30,000

Asset 3: This machine was acquired by making a €10,000 down payment and issuing a €30,000, 2-year, zero-interest-bearing note. The note is to be paid off in two €15,000 installments made at the end of the first and second years. It was estimated that the asset could have been purchased outright for €35,900.

Asset 4: This machinery was acquired by trading in used machinery. (The exchange lacks commercial substance.) Facts concerning the trade-in are as follows.

Cost of machinery traded	€100,000
Accumulated depreciation to date of sale	36,000
Fair value of machinery traded	80,000
Cash received	10,000
Fair value of machinery acquired	70,000

Asset 5: Equipment was acquired by issuing 100 shares of €8 par value ordinary shares. The shares have a market price of €11 per share.

Construction of Building: A building was constructed on land with a cost of €180,000. Construction began on February 1 and was completed on November 1. The payments to the contractor were as follows.

Date	Payment
2/1	€120,000
6/1	360,000
9/1	480,000
11/1	100,000

To finance construction of the building, a €600,000, 12% construction loan was taken out on February 1. The loan was repaid on November 1. The firm had

€200,000 of other outstanding debt during the year at a borrowing rate of 8%.

Instructions

Record the acquisition of each of these assets. Interest expense for the year has been recorded.

E9.17 (LO 3) (Non-Monetary Exchange) Alatorre SpA, which manufactures shoes, hired a recent college graduate to work in its accounting department. On the first day of work, the accountant was assigned to total a batch of invoices with the use of an adding machine. Before long, the accountant, who had never before seen such a machine, broke it. Alatorre gave the machine plus €320 to Mills Business Machine AG (dealer) in exchange for a new machine. Assume the following information about the machines.

	Alatorre SpA (Old Machine)	Mills AG (New Machine)
Machine cost	€290	€270
Accumulated depreciation	140	–0–
Fair value	85	405

Instructions

For each company, prepare the necessary journal entry to record the exchange. (The exchange has commercial substance.)

E9.18 (LO 3) (Non-Monetary Exchange) Montgomery Ltd. purchased an electric wax melter on April 30, 2026, by trading in its old gas model and paying the balance in cash. The following data relate to the purchase.

List price of new melter	£15,800
Cash paid	10,000
Cost of old melter (5-year life, £700 residual value)	12,700
Accumulated depreciation (old melter—straight-line)	7,200
Fair value of old melter	5,200

Instructions

Prepare the journal entry or entries necessary to record this exchange, assuming that the exchange (a) has commercial substance, and (b) lacks commercial substance. Montgomery's year ends on December 31, and depreciation has been recorded through December 31, 2025.

E9.19 (LO 3) (Non-Monetary Exchange) Santana SA exchanged equipment used in its manufacturing operations plus R$2,000 in cash for similar equipment used in the operations of Delaware Co. The following information pertains to the exchange.

	Santana SA	Delaware Co.
Equipment (cost)	R$28,000	R$28,000
Accumulated depreciation	19,000	10,000
Fair value of equipment	13,500	15,500
Cash given up	2,000	

Instructions

a. Prepare the journal entries to record the exchange on the books of both companies. Assume that the exchange lacks commercial substance.

b. Prepare the journal entries to record the exchange on the books of both companies. Assume that the exchange has commercial substance.

E9.20 (LO 3) (Non-Monetary Exchange) Yintang Group has negotiated the purchase of a new piece of automatic equipment at a price of HK$7,000 plus trade-in, f.o.b. factory. Yintang paid HK$7,000 cash and traded in used equipment. The used equipment had originally cost HK$62,000; it had a book value of HK$42,000 and a secondhand fair value of HK$45,800, as indicated by recent transactions involving similar equipment. Freight and installation charges for the new equipment required a cash payment of HK$1,100.

Instructions

a. Prepare the general journal entry to record this transaction, assuming that the exchange has commercial substance.

b. Assuming the same facts as in (a) except that fair value information for the assets exchanged is not determinable, prepare the general journal entry to record this transaction.

E9.21 (LO 3) (Government Grants) Rialto Group received a £100,000 grant from the government to acquire £500,000 of delivery equipment on January 2, 2025. The delivery equipment has a useful life of 5 years. Rialto uses the straight-line method of depreciation. The delivery equipment has a zero residual value.

Instructions

a. If Rialto Group reports the grant as a reduction of the asset, answer the following questions.

   1. What is the carrying amount of the delivery equipment on the statement of financial position at December 31, 2025?

2. What is the amount of depreciation expense related to the delivery equipment in 2026?
3. What is the amount of grant revenue reported in 2025 on the income statement?

b. If Rialto Group reports the grant as deferred grant revenue, answer the following questions.

1. What is the balance in the deferred grant revenue account at December 31, 2025?
2. What is the amount of depreciation expense related to the delivery equipment in 2026?
3. What is the amount of grant revenue reported in 2025 on the income statement?

E9.22 **(LO 3)** **(Government Grants)** Yilmaz SA is provided a grant by the local government to purchase land for a building site. The grant is a zero-interest-bearing note for 5 years. The note is issued on January 2, 2025, for €5 million payable on January 2, 2030. Yilmaz's incremental borrowing rate is 6%. The land is not purchased until July 15, 2025.

Instructions

a. Prepare the journal entry or entries to record the grant and note payable on January 2, 2025.

b. Determine the amount of interest expense and grant revenue to be reported on December 31, 2025.

E9.23 **(LO 4)** Groupwork **(Analysis of Subsequent Expenditures)** Accardo Resources Group has been in its plant facility for 15 years. Although the plant is quite functional, numerous repair costs are incurred to maintain it in sound working order. The company's plant asset book value is currently R$800,000, as indicated below.

Original cost	R$1,200,000
Less: Accumulated depreciation	400,000
Book value	R$ 800,000

During the current year, the following improvements were made to the plant facility.

a. Because of increased demands for its product, the company increased its plant capacity by building a new addition at a cost of R$270,000.

b. The entire plant was repainted at a cost of R$23,000.

c. The original roof was an asbestos cement slate. For safety purposes, it was removed and replaced with a wood shingle roof at a cost of R$61,000. Book value of the old roof was R$41,000.

d. The electrical system was updated at a cost of R$12,000. The cost of the old electrical system was not known. It is estimated that the useful life of the building will not change as a result of this updating.

e. A series of major repairs were made at a cost of R$75,000 because parts of the wood structure were rotting. The cost of the old wood structure was not known. These extensive repairs are estimated to increase the useful life of the building. The company believes the R$75,000 is representative of the cost of parts for the wood structure at the date of purchase.

Instructions

Indicate how each of these transactions would be recorded in the accounting records.

E9.24 (LO 4) (Analysis of Subsequent Expenditures) The following transactions occurred during 2026. Assume that depreciation of 10% per year is charged on all machinery and 5% per year on buildings, on a straight-line basis, with no estimated residual value. Depreciation is charged for a full year on all fixed assets acquired during the year, and no depreciation is charged on fixed assets disposed of during the year.

Jan. 30	A building that cost $112,000 in 2009 is torn down to make room for a new building. The wrecking contractor was paid $5,100 and was permitted to keep all materials salvaged.
Mar. 10	Machinery that was purchased in 2019 for $16,000 is sold for $2,900 cash, f.o.b. purchaser's plant. Freight of $300 is paid on the sale of this machinery.
Mar. 20	A gear breaks on a machine that cost $9,000 in 2021. The gear is replaced at a cost of $3,000. The replacement does not extend the useful life of the machine.
May 18	A special base installed for a machine in 2020 when the machine was purchased has to be replaced at a cost of $5,500 because of defective workmanship on the original base. The cost of the machinery was $14,200 in 2020. The cost of the base was $4,000, and this amount was charged to the Machinery account in 2020.
June 23	One of the buildings is repainted at a cost of $6,900. It had not been painted since it was constructed in 2022.

Instructions

Prepare general journal entries for the transactions. (Round to the nearest dollar.)

E9.25 (LO 4) (Analysis of Subsequent Expenditures) Plant assets often require expenditures subsequent to acquisition. It is important that they be accounted for properly. Any errors will affect both the statements of financial position and income statements for a number of years.

Instructions

For each of the following items, indicate whether the expenditure should be capitalized (C) or expensed (E) in the period incurred.

a. ________	Improvement.
b. ________	Replacement of a minor broken part on a machine.
c. ________	Expenditure that increases the useful life of an existing asset.
d. ________	Expenditure that increases the efficiency and effectiveness of a productive asset but does not increase its residual value.
e. ________	Expenditure that increases the efficiency and effectiveness of a productive asset and increases the asset's residual value.
f. ________	Ordinary repairs.
g. ________	Improvement to a machine that increased its fair value and its production capacity by 30% without extending the machine's useful life.
h. ________	Expenditure that increases the quality of the output of the productive asset.

E9.26 (LO 5) (Entries for Disposition of Assets) On December 31, 2025, Mitsui Ltd. has a machine with a book value of ¥940,000. The original cost and related accumulated depreciation at this date are as follows (all amounts in thousands).

Machine	¥1,300,000
Less: Accumulated depreciation	360,000
Book value	¥ 940,000

Depreciation is computed at ¥72,000 per year on a straight-line basis.

Instructions

Presented below is a set of independent situations. For each independent situation, indicate the journal entry to be made to record the transaction. Make sure that depreciation entries are made to update the book value of the machine prior to its disposal.

a. A fire completely destroys the machine on August 31, 2023. An insurance settlement of ¥630,000 was received for this casualty. The insurance company confirmed on September 30 that the claim would be covered and that the amount of the settlement (¥630,000) would be paid on October 15.

b. On April 1, 2023, Mitsui sold the machine for ¥1,040,000 to Avanti Company.

c. On July 31, 2023, the company donated this machine to the Mountain King City Council. The fair value of the machine at the time of the donation was estimated to be ¥1,100,000.

E9.27 (LO 5) (Disposition of Assets) On January 1, 2025, Pavlova Company received a notice from the government that a new highway was being built and that land and a building owned by the company were going to be claimed in the process. The company employed an appraiser and estimated that the land had a fair value of $114,000 and the building had a fair value of $385,000. On April 30, 2025, after some negotiations, Pavlova received a condemnation award of $520,000 cash as compensation for the forced sale of the company's land and building. The land and building cost $120,000 and $600,000, respectively, when they were acquired. On January 1, 2025, the accumulated depreciation relating to the building amounted to $208,000. On April 30, the accumulated depreciation relating to the building was $210,000. On August 1, 2025, Pavlova purchased a piece of replacement property for cash. The new land cost $90,000, and the new building cost $380,000.

Instructions

Prepare the journal entries to record the transactions on January 1, April 30, and August 1, 2025.

Problems

P9.1 (LO 1, 2) (Classification of Acquisition and Other Asset Costs) At December 31, 2024, certain accounts included in the property, plant, and equipment section of Reagan Ltd.'s statement of financial position had the following balances.

Land	£230,000
Buildings	890,000

Leasehold improvements	660,000
Machinery and equipment	875,000

During 2025, the following transactions occurred.

1. Land site number 621 was acquired for £850,000. In addition, to acquire the land Reagan paid a £51,000 commission to a real estate agent. Costs of £35,000 were incurred to clear the land. During the course of clearing the land, timber and gravel were recovered and sold for £13,000.

2. A second tract of land (site number 622) with a building was acquired for £420,000. The closing statement indicated that the land value was £300,000 and the building value was £120,000. Shortly after acquisition, the building was demolished at a cost of £41,000. A new building was constructed for £330,000 plus the following costs.

Excavation fees	£38,000
Architectural design fees	11,000
Building permit fee	2,500
Imputed interest on funds used during construction (share financing)	8,500

   The building was completed and occupied on September 30, 2025.

3. A third tract of land (site number 623) was acquired for £650,000 and was put on the market for resale.

4. During December 2025, costs of £89,000 were incurred to improve leased office space. The related lease will terminate on December 31, 2027, and is not expected to be renewed. (*Hint:* Leasehold improvements should be handled in the same manner as land improvements.)

5. A group of new machines was purchased under a royalty agreement that provides for payment of royalties based on units of production for the machines. The invoice price of the machines was £87,000, freight costs were £3,300, installation costs were £2,400, and royalty payments for 2025 were £17,500.

Instructions

a. Prepare a detailed analysis of the changes in each of the following statement of financial position accounts for 2025.

Land	Leasehold improvements
Buildings	Machinery and equipment

Disregard the related accumulated depreciation accounts.

b. List the items that were not used to determine the answer to (a) above, and indicate where, or if, these items should be included in Reagan's financial statements.

P9.2 (LO 1, 5) (Classification of Acquisition Costs) Selected accounts included in the property, plant, and equipment section of Lobo Corporation's statement of financial position at December 31, 2024, had the following balances.

Land	$ 300,000
Land improvements	140,000
Buildings	1,100,000
Equipment	960,000

During 2024, the following transactions occurred.

1. A tract of land was acquired for $150,000 as a potential future building site.
2. A plant facility consisting of land and building was acquired from Mendota Company in exchange for 20,000 shares of Lobo's ordinary shares. On the acquisition date, Lobo's shares had a closing market price of $37 per share on a national exchange. The plant facility was carried on Mendota's books at $110,000 for land and $320,000 for the building at the exchange date. Current appraised values for the land and building, respectively, are $230,000 and $690,000.
3. Items of equipment were purchased at a total cost of $400,000. Additional costs were incurred as follows.

Freight and unloading	$13,000
Sales taxes	20,000
Installation	26,000

4. Expenditures totaling $95,000 were made for new parking lots, streets, and sidewalks at the company's various plant locations. These expenditures had an estimated useful life of 15 years.
5. Equipment costing $80,000 on January 1, 2017, was scrapped on June 30, 2025. Straight-line depreciation has been recorded on the basis of a 10-year life with no residual values.
6. Equipment was sold for $20,000 on July 1, 2025. Original cost of the equipment was $44,000 on January 1, 2022, and it was depreciated on the straight-line basis over an estimated useful life of 7 years and a residual value of $2,000.

Instructions

a. Prepare a detailed analysis of the changes in each of the following statement of financial position accounts for 2025.

Land	Buildings
Land improvements	Equipment

(*Hint:* Disregard the related accumulated depreciation accounts.)

b. List the items that were not used to determine the answer to (a), showing the pertinent amounts and supporting computations in good form for each item. In addition, indicate where, or if, these items should be included in Lobo's financial statements.

P9.3 (LO 1, 3) (Classification of Land and Building Costs) Spitfire SE started as a company on January 2, 2026, but was unable to begin manufacturing activities until July 1, 2026, because new factory facilities were not completed until that date.

The Land and Buildings account reported the following items during 2026.

Jan. 31	Land and building	€160,000
Feb. 28	Cost of removal of building	9,800
May 1	Partial payment of new construction	60,000
May 1	Legal fees paid	3,770
June 1	Second payment on new construction	40,000
June 1	Insurance premium	2,280
June 1	Special tax assessment	4,000
June 30	General expenses	36,300
July 1	Final payment on new construction	30,000
Dec. 31	Asset write-up	53,800
		399,950
Dec. 31	Depreciation—2026 at 1%	4,000
Dec. 31, 2026	Account balance	€395,950

The following additional information is to be considered.

1. To acquire land and building, the company paid €80,000 cash and 800 shares of its 8% preference shares, par value €100 per share. The shares trade in an active market at €117 per share.
2. Cost of removal of old buildings amounted to €9,800, and the demolition company retained all materials of the building.

3. Legal fees covered the following.

Cost of organization	€ 610
Examination of title covering purchase of land	1,300
Legal work in connection with construction contract	1,860
	€3,770

4. Insurance premium covered the building for a 2-year term beginning May 1, 2026.
5. The special tax assessment covered street improvements that are permanent in nature.
6. General expenses covered the following for the period from January 2, 2026, to June 30, 2026.

President's salary	€32,100
Plant superintendent's salary—supervision of new building	4,200
	€36,300

7. Because of a general increase in construction costs after entering into the building contract, the board of directors increased the value of the building €53,800, believing that such an increase was justified to reflect the current market at the time the building was completed. Retained earnings was credited for this amount.
8. Depreciation for 2026—1% of asset value (1% of €400,000, or €4,000).

Instructions

a. Prepare entries to reflect correct land, buildings, and depreciation accounts at December 31, 2026.

b. Show the proper presentation of land, buildings, and depreciation on the statement of financial position at December 31, 2026.

P9.4 (LO 1, 3, 5) Groupwork **(Dispositions, Including Condemnation, Demolition, and Trade-In)** Presented below is a schedule of property dispositions for Hollerith Co.

Schedule of Property Dispositions					
	Cost	Accumulated Depreciation	Cash Proceeds	Fair Value	Nature of Disposition
Land	$40,000	—	$31,000	$31,000	Condemnation
Building	15,000	—	3,600	—	Demolition

Warehouse	70,000	$16,000	74,000	74,000	Destruction by fire
Machine	8,000	2,800	900	7,200	Trade-in
Furniture	10,000	7,850	—	3,100	Contribution
Automobile	9,000	3,460	2,960	2,960	Sale

The following additional information is available.

Land: On February 15, a condemnation award was received as consideration for unimproved land held primarily as an investment. On March 31, another parcel of unimproved land to be held as an investment was purchased at a cost of $35,000. The company had been made aware that the government was going to exercise its right of eminent domain in the previous year and had written down the land to its estimated fair value of $25,000 on December 31.

Building: On April 2, land and building were purchased at a total cost of $75,000, of which 20% was allocated to the building on the company books. The real estate was acquired with the intention of demolishing the building, and this was accomplished during the month of November. Cash proceeds received in November represent the net proceeds from demolition of the building.

Warehouse: On June 30, the warehouse was destroyed by fire. The insurance company agreed to cover the claim and disbursed the $74,000 on July 31. The warehouse was purchased January 2, 2019, and had depreciated $16,000. On December 27, the insurance proceeds and other funds were used to purchase a replacement warehouse at a cost of $90,000.

Machine: On December 26, the machine was exchanged for another machine having a fair value of $6,300 and cash of $900 was received. (The exchange lacks commercial substance.)

Furniture: On August 15, furniture was contributed to a qualified charitable organization. No other contributions were made or pledged during the year.

Automobile: On November 3, the automobile was sold to Jared Winger, a shareholder.

Instructions

Indicate how these items would be reported on the income statement of Hollerith Co.

P9.5 (LO 1, 2) (Classification of Costs and Capitalization of Borrowing Costs)
On January 1, 2025, Blair Corporation purchased for $500,000 a tract of land (site

number 101) with a building. Blair paid a real estate broker's commission of $36,000, legal fees of $6,000, and title guarantee insurance of $18,000. The closing statement indicated that the land value was $500,000 and the building value was $100,000. Shortly after acquisition, the building was razed at a cost of $54,000.

Blair entered into a $3,000,000 fixed-price contract with Slatkin Builders, Inc. on March 1, 2025, for the construction of an office building on land site number 101. The building was completed and occupied on September 30, 2026. Additional construction costs were incurred as follows.

Plans, specifications, and blueprints	$21,000
Architects' fees for design and supervision	82,000

The building is estimated to have a 40-year life from date of completion and will be depreciated using the 150% declining-balance method.

To finance construction costs, Blair borrowed $3,000,000 at a 10% annual interest rate on March 1, 2025. Blair will pay interest only over the life of the loan, with the principal payable at the end of 10 years when the loan matures. Excess funds from the construction loan were invested to earn a return. During 2025, these funds earned interest revenue of $135,000. During 2026, the funds earned interest revenue of $45,000.

Instructions

a. Prepare a schedule that discloses the individual costs making up the balance in the Land account in respect of land site number 101 as of September 30, 2026.

b. Prepare a schedule that discloses the individual costs that should be capitalized in the Buildings account as of September 30, 2026. Show supporting computations in good form.

P9.6 (LO 2) (Borrowing Costs) Cho Landscaping began construction of a new plant on December 1, 2025 (all amounts in thousands). On this date, the company purchased a parcel of land for ¥139,000 in cash. In addition, it paid ¥2,000 in surveying costs and ¥4,000 for a title insurance policy. An old dwelling on the premises was demolished at a cost of ¥3,000, with ¥1,000 being received from the sale of materials.

Architectural plans were also formalized on December 1, 2025, when the architect was paid ¥30,000. The necessary building permits costing ¥3,000 were obtained from the city and paid for on December 1 as well. The excavation work began during the first week in December with payments made to the contractor in 2026 as follows.

Date of Payment	Amount of Payment
March 1	¥240,000
May 1	330,000
July 1	60,000

The building was completed on July 1, 2026.

To finance construction of this plant, Cho borrowed ¥600,000 from the bank on December 1, 2025. Cho had no other borrowings. The ¥600,000 was a 10-year loan bearing interest at 8%. Excess funds from the loan were invested during 2025 and earned ¥2,000. During 2026, the excess funds from the loan earned ¥1,000.

Instructions

Compute the balance in each of the following accounts at December 31, 2025, and December 31, 2026. (Round amounts to the nearest 1,000 yen.)

a. Land.

b. Buildings.

c. Interest Expense.

P9.7 (LO 2) Groupwork (Capitalization of Borrowing Costs) Laserwords A.Ş. is a book distributor that had been operating in its original facility since 1988. The increase in certification programs and continuing education requirements in several professions has contributed to an annual growth rate of 15% for Laserwords since 2020. Laserwords' original facility became obsolete by early 2025 because of the increased sales volume and the fact that Laserwords now carries CDs in addition to books.

On June 1, 2025, Laserwords contracted with Black Construction to have a new building constructed for ₺4,000,000 on land owned by Laserwords. The payments made by Laserwords to Black Construction are shown in the schedule below.

Date	Amount
July 30, 2025	₺ 900,000
January 30, 2026	1,500,000
May 30, 2026	1,600,000
Total payments	₺ 4,000,000

Construction was completed, and the building was ready for occupancy on May 27, 2026. Laserwords had no new borrowings directly associated with the new building but had the following debt outstanding at May 31, 2026, the end of its fiscal year.

10%, 5-year note payable of ₺2,000,000, dated April 1, 2022, with interest payable annually on April 1.

12%, 10-year bond issue of ₺3,000,000 sold at par on June 30, 2018, with interest payable annually on June 30.

The new building qualifies for capitalization of borrowing costs. The effect of capitalizing the interest on the new building, compared with the effect of expensing the interest, is material.

Instructions

a. Compute the average carrying amount on Laserwords' new building during the capitalization period.

b. Compute the borrowing cost to be capitalized on Laserwords' new building.

c. Some borrowing costs of Laserwords Inc. are capitalized for the year ended May 31, 2026.

   1. Identify the items relating to borrowing costs that must be disclosed in Laserwords' financial statements.

   2. Compute the amount of each of the items that must be disclosed.

P9.8 **(LO 3)** **(Non-Monetary Exchanges)** Holyfield SA wishes to exchange a machine used in its operations. Holyfield has received the following offers from other companies in the industry.

1. Dorsett Company offered to exchange a similar machine plus €23,000. (The exchange has commercial substance for both parties.)

2. Winston Company offered to exchange a similar machine. (The exchange lacks commercial substance for both parties.)

3. Liston Company offered to exchange a similar machine but wanted €3,000 in addition to Holyfield's machine. (The exchange has commercial substance for both parties.)

4. In addition, Holyfield contacted Greeley Group, a dealer in machines. To obtain a new machine, Holyfield must pay €93,000 in addition to trading in its old machine. (The exchange has commercial substance.)

	Holyfield	Dorsett	Winston	Liston	Greeley
Machine cost	€160,000	€120,000	€152,000	€160,000	€130,000
Accumulated depreciation	60,000	45,000	71,000	75,000	–0–
Fair value	92,000	69,000	92,000	95,000	185,000

Instructions

For each of the four independent situations, prepare the journal entries to record the exchange on the books of each company.

P9.9 (LO 3) (Non-Monetary Exchanges) On August 1, Hyde Ltd. exchanged productive assets with Wiggins Ltd. Hyde's asset is referred to below as Asset A, and Wiggins' is referred to as Asset B. The following facts pertain to these assets.

	Asset A	Asset B
Original cost	£96,000	£110,000
Accumulated depreciation (to date of exchange)	40,000	47,000
Fair value at date of exchange	60,000	75,000
Cash paid by Hyde Ltd.	15,000	
Cash received by Wiggins Ltd.		15,000

Instructions

a. Assuming that the exchange of Assets A and B has commercial substance, record the exchange for both Hyde Ltd. and Wiggins Ltd.

b. Assuming that the exchange of Assets A and B lacks commercial substance, record the exchange for both Hyde Ltd. and Wiggins Ltd.

P9.10 (LO 3) (Non-Monetary Exchanges) During the current year, Marshall Construction trades an old crane that has a book value of €90,000 (original cost €140,000 less accumulated depreciation €50,000) for a new crane from Brigham Manufacturing. The new crane cost Brigham €165,000 to manufacture and is classified as inventory. The following information is also available.

	Marshall Const.	Brigham Mfg.
Fair value of old crane	€ 82,000	
Fair value of new crane		€200,000
Cash paid	118,000	
Cash received		118,000

Instructions

a. Assuming that this exchange is considered to have commercial substance, prepare the journal entries on the books of (1) Marshall Construction and (2) Brigham Manufacturing.

b. Assuming that this exchange lacks commercial substance for Marshall, prepare the journal entries on the books of Marshall Construction.

c. Assuming the same facts as those in (a) except that the fair value of the old crane is €98,000 and the cash paid is €102,000, prepare the journal entries on the books of (1) Marshall Construction and (2) Brigham Manufacturing.

d. Assuming the same facts as those in (b) except that the fair value of the old crane is €97,000 and the cash paid €103,000, prepare the journal entries on the books of (1) Marshall Construction and (2) Brigham Manufacturing.

P9.11 (LO 1, 3, 5) (Purchases by Deferred Payment, Lump-Sum, and Non-Monetary Exchanges) Kang Ltd., a manufacturer of ballet shoes, is experiencing a period of sustained growth. In an effort to expand its production capacity to meet the increased demand for its product, the company recently made several acquisitions of plant and equipment. Jacky Cheung, newly hired in the position of fixed-asset accountant, requested that Tanya Chan,, Kang's controller, review the following transactions.

Transaction 1: On June 1, 2025, Kang Ltd. purchased equipment from Wyandot Group. Kang issued a HK$28,000, 4-year, zero-interest-bearing note to Wyandot for the new equipment. Kang will pay off the note in four equal installments due at the end of each of the next 4 years. At the date of the transaction, the prevailing market rate of interest for obligations of this nature was 10%. Freight costs of HK$425 and installation costs of HK$500 were incurred in completing this transaction. The appropriate factors for the time value of money at a 10% rate of interest are given below.

Future value of HK$1 for 4 periods	1.46
Future value of an ordinary annuity for 4 periods	4.64
Present value of HK$1 for 4 periods	0.68
Present value of an ordinary annuity for 4 periods	3.17

Transaction 2: On December 1, 2025, Kang Ltd. purchased several assets of Yakima Shoes, a small shoe manufacturer whose owner was retiring. The purchase amounted to HK$220,000 and included the assets listed below. Kang engaged the services of Tennyson Appraisal, an independent appraiser, to determine the fair values of the assets which are also presented below.

	Yakima Book Value	Fair Value
Inventory	HK$ 60,000	HK$ 50,000
Land	40,000	80,000
Buildings	70,000	120,000
	HK$170,000	HK$250,000

During its fiscal year ended May 31, 2026, Kang incurred HK$8,000 for interest expense in connection with the financing of these assets.

Transaction 3: On March 1, 2026, Kang Ltd. exchanged a number of used trucks plus cash for vacant land adjacent to its plant site. (The exchange has commercial substance.) Kang intends to use the land for a parking lot. The trucks had a combined book value of HK$35,000, as Kang had recorded HK$20,000 of accumulated depreciation against these assets. Kang's purchasing agent, who has had previous dealings in the secondhand market, indicated that the trucks had a fair value of HK$46,000 at the time of the transaction. In addition to the trucks, Kang paid HK$19,000 cash for the land.

Instructions

a. Plant assets such as land, buildings, and equipment receive special accounting treatment. Describe the major characteristics of these assets that differentiate them from other types of assets.

b. For each of the three transactions described above, determine the value at which Kang Ltd. should record the acquired assets. Support your calculations with an explanation of the underlying rationale.

c. The books of Kang Ltd. show the following additional transactions for the fiscal year ended May 31, 2026.

   1. Acquisition of a building for speculative purposes.
   2. Purchase of a 2-year insurance policy covering plant equipment.
   3. Purchase of the rights for the exclusive use of a process used in the manufacture of ballet shoes.

For each of these transactions, indicate whether the asset should be classified as a plant asset. If it is a plant asset, explain why it is. If it is not a plant asset, explain why not, and identify the proper classification.

Concepts for Analysis

CA9.1 (LO 1, 4, 5) Writing (Acquisition, Improvements, and Sale of Realty) Tonkawa Group purchased land for use as its company headquarters. A small factory that was on the land when it was purchased was torn down before construction of the office building began. Furthermore, a substantial amount of rock blasting and removal had to be done before construction of the building foundation began. Because the office building was set back on the land far from the public road, Tonkawa Group had the contractor construct a paved road that led from the public road to the parking lot of the office building.

Three years after the office building was occupied, Tonkawa Group added four stories to the office building. The four stories had an estimated useful life of 5 years more than the remaining estimated useful life of the original office building.

Ten years later, the land and building were sold at an amount more than their net book value, and Tonkawa had a new office building constructed at another site for use as its new company headquarters.

Instructions

a. Which of the expenditures above should be capitalized? How should each be depreciated or amortized? Discuss the rationale for your answers.

b. How would the sale of the land and building be accounted for? Include in your answer an explanation of how to determine the net book value at the date of sale. Discuss the rationale for your answer.

CA9.2 (LO 1) (Accounting for Self-Constructed Assets) Troopers Medical Labs began operations 5 years ago producing stetrics, a new type of instrument it hoped to sell to doctors, dentists, and hospitals. The demand for stetrics far exceeded initial expectations, and the company was unable to produce enough to meet demand.

The company was manufacturing its product on equipment that it built at the start of its operations. To meet demand, more efficient equipment was needed. The company decided to design and build the equipment because the equipment currently available on the market was unsuitable for producing stetrics.

In 2025, a section of the plant was devoted to development of the new equipment and a special staff was hired. Within 6 months, a machine developed at a cost of €714,000 increased production dramatically and reduced labor costs substantially. Elated by the success of the new machine, the company built three more machines of the same type at a cost of €441,000 each.

Instructions

a. In general, what costs should be capitalized for self-constructed equipment?

b. Discuss the propriety of including in the capitalized cost of self-constructed assets:

   1. The increase in overhead caused by the self-construction of fixed assets.
   2. A proportionate share of overhead on the same basis as that applied to goods manufactured for sale.

c. Discuss the proper accounting treatment of the €273,000 (€714,000 – €441,000) by which the cost of the first machine exceeded the cost of the subsequent machines. This additional cost should not be considered a research and development cost.

CA9.3 **(LO 2)** **(Capitalization of Borrowing Costs)** Langer Airline is converting from piston-type planes to jets. Delivery time for the jets is 3 years, during which time substantial progress payments must be made. The multimillion-dollar cost of the planes cannot be financed from working capital; Langer must borrow funds for the payments.

Because of high interest rates and the large sum to be borrowed, management estimates that borrowing costs in the second year of the period will be equal to one-third of income before interest and taxes, and one-half of such income in the third year.

After conversion, Langer's passenger-carrying capacity will be doubled with no increase in the number of planes, although the investment in planes will be substantially increased. The jet planes have a 7-year service life.

Instructions

Give your recommendation concerning the proper accounting for interest during the conversion period. Support your recommendation with reasons and suggested accounting treatment. (Disregard income tax implications.)

CA9.4 **(LO 2)** **Writing** **(Capitalization of Borrowing Costs)** Vang Magazine Group started construction of a warehouse building for its own use at an estimated cost of ¥5,000,000 on January 1, 2024, and completed the building on December 31, 2024 (all amounts in thousands). During the construction period, Vang has the following debt obligations outstanding.

Construction loan—12% interest, payable semiannually, issued December 31, 2023	¥2,000,000
Short-term loan—10% interest, payable monthly, and principal payable at maturity, on May 30, 2025	1,400,000
Long-term loan—11% interest, payable on January 1 of each year. Principal payable on January 1, 2027	1,000,000

Total cost amounted to ¥5,200,000. Payments were made during the year as follows.

Date	Amount
January 1, 2025	¥1,500,000
March 1, 2025	1,250,000
September 1, 2025	1,250,000

December 1, 2025	1,200,000
Total	¥5,200,000

Jane Edo, the president of the company, has been shown the costs associated with this construction project and capitalized on the statement of financial position. She is bothered by the "borrowing cost." She argues that, first, all the interest is unavoidable—no one lends money without expecting to be compensated for it. Second, why can't the company use all the interest on all the loans when computing the capitalized borrowing cost? Finally, why can't her company capitalize all the annual interest that accrued over the period of construction?

Instructions

You are the manager of accounting for the company. In a memo dated January 15, 2026, explain what borrowing cost is, how you computed it (being especially careful to explain why you used the interest rates that you did), and why the company cannot capitalize all its interest for the year. Attach a schedule supporting any computations that you use.

CA9.5 (LO 3) Writing **(Non-Monetary Exchanges)** You have two clients who are considering trading machinery with each other. Although the machines are different from each other, you believe that an assessment of expected cash flows on the exchanged assets will indicate the exchange lacks commercial substance. Your clients would prefer that the exchange be deemed to have commercial substance, to allow them to record gains. Here are the facts:

	Client A	Client B
Original cost	£100,000	£150,000
Accumulated depreciation	40,000	80,000
Fair value	80,000	100,000
Cash received (paid)	(20,000)	20,000

Instructions

a. Record the trade-in on Client A's books assuming the exchange has commercial substance.

b. Record the trade-in on Client A's books assuming the exchange lacks commercial substance.

c. Write a memo to the controller of Company A indicating and explaining the specific financial impact on current and future statements of treating the exchange as having versus lacking commercial substance.

d. Record the entry on Client B's books assuming the exchange has commercial substance.

e. Record the entry on Client B's books assuming the exchange lacks commercial substance.

f. Write a memo to the controller of Company B indicating and explaining the specific financial impact on current and future statements of treating the exchange as having versus lacking commercial substance.

CA9.6 (LO 1) **(Costs of Acquisition)** The invoice price of a machine is €50,000. Various other costs relating to the acquisition and installation of the machine, including transportation, electrical wiring, special base, and so on, amount to €7,500. The machine has an estimated life of 10 years, with no residual value at the end of that period.

The owner of the business suggests that the incidental costs of €7,500 be charged to expense immediately for the following reasons.

1. If the machine should be sold, these costs cannot be recovered in the sales price.
2. The inclusion of the €7,500 in the machinery account on the books will not necessarily result in a closer approximation of the market price of this asset over the years because of the possibility of changing demand and supply levels.
3. Charging the €7,500 to expense immediately will reduce income taxes.

Instructions

Discuss each of the points raised by the owner of the business.

CA9.7 (LO 1) Ethics **(Cost of Land vs. Building—Ethics)** Tones Company purchased a warehouse in a downtown district where land values are rapidly increasing. Gerald Carter, controller, and Wilma Ankara, financial vice president, are trying to allocate the cost of the purchase between the land and the building. Noting that depreciation can be taken only on the building, Carter favors placing a very high proportion of the cost on the warehouse itself, thus reducing taxable income and income taxes. Ankara, his supervisor, argues that the allocation should recognize the increasing value of the land, regardless of the depreciation potential of the warehouse. Besides, she says, net income is negatively impacted by additional depreciation and will cause the company's share price to go down.

Instructions

Answer the following questions.

a. What stakeholder interests are in conflict?

b. What ethical issues does Carter face?

c. How should these costs be allocated?

Using Your Judgment

Financial Statement Analysis Case

Unilever Group

Unilever Group (GBR) is a leading international company in the nutrition, hygiene, and personal-care product lines. Information related to Unilever's property, plant, and equipment in its 2018 annual report is shown in the notes to the financial statements as follows.

10. Property, Plant and Equipment

Property, plant and equipment is measured at cost including eligible borrowing costs less depreciation and accumulated impairment losses. Depreciation is provided on a straight-line basis over the expected average useful lives of the assets. Residual values are reviewed at least annually.

Estimated useful lives by major class of assets are as follows.

• Freehold buildings [no depreciation on freehold land]	40 years
• Leasehold land and buildings	40 years [or life of lease if less]
• Plant and equipment	2–20 years

Property, plant and equipment is subject to review for impairment if triggering events or circumstances indicate that this is necessary. If an indication of impairment exists, the asset or cash generating unit recoverable amount is estimated and any impairment loss is charged to the income statement as it arises.

Movements during 2018	€ million Land and buildings	€ million Plant and equipment	€ million Total
Cost			
1 January 2018	4,462	14,936	19,398
Hyperinflation restatement to 1 January 2018	37	182	219

Acquisitions of group companies	11	31	42
Additions	249	1,091	1,340
Disposals	(97)	(607)	(704)
Hyperinflationary adjustment	49	93	142
Currency retranslation	(91)	(351)	(442)
Reclassification as held for sale	(17)	(54)	(71)
31 December 2018	4,603	15,321	19,924
Accumulated depreciation			
1 January 2018	(1,429)	(7,558)	(8,987)
Hyperinflation restatement to 1 January 2018	(10)	(106)	(116)
Depreciation charge for the year	(125)	(1,066)	(1,191)
Disposals	62	529	591
Hyperinflationary adjustment	(7)	(53)	(60)
Currency retranslation	15	128	143
Reclassification as held for sale	10	33	43
31 December 2018	(1,484)	(8,093)	(9,577)
Net book value 31 December 2018	3,119	7,228	10,347
Includes capital expenditure for assets under construction	130	956	1,086

The Group has committed capital expenditure of €324 million (2017: €323 million).

Unilever provided the following selected information in its 2018 cash flow statement.

Unilever (€ million)	
	€ million 2018
Net profit	9,808
Taxation	2,575

Share of net profit of joint ventures/associates and other income/(loss) from non-current investments and associates	(207)
Net monetary gain arising from hyperinflationary economies	(122)
Net finance costs	481
Operating profit	12,535
Depreciation, amortization and impairment	1,747
Changes in working capital:	(793)
Inventories	(471)
Trade and other receivables	(1,298)
Trade payables and other liabilities	976
Pensions and similar obligations less payments	(128)
Provisions less payments	55
Elimination of (profits)/losses on disposals	(4,299)
Non-cash charge for share-based compensation	196
Other adjustments	(266)
Cash flow from operating activities	9,047
Income tax paid	(2,294)
Net cash flow from operating activities	6,753
Interest received	110
Purchase of intangible assets	(203)
Purchase of property, plant, and equipment	(1,329)
Disposal of property, plant, and equipment	108
Acquisition of group companies, joint ventures, and associates	(1,336)
Disposal of group companies, joint ventures, and associates	7,093
Acquisition of other non-current investments	(94)
Disposal of other non-current investments	151
Dividends from joint ventures, associates, and other non-current investments	154
(Purchase)/sale of financial assets	(10)
Net cash flow (used in)/from investing activities	4,644
Dividends paid on ordinary share capital	(4,066)
Interest and preference dividends paid	(477)

Net change in short-term borrowings	(4,026)
Additional financial liabilities	10,595
Repayment of financial liabilities	(6,594)
Capital element of finance lease rental payments	(10)
Buyback of preference shares	–
Repurchase of shares	(6,020)
Other movements on treasury shares	(257)
Other financing activities	(693)
Net cash flow (used in)/from financing activities	(11,548)
Net increase/(decrease) in cash and cash equivalents	(151)
Cash and cash equivalents at the beginning of the year	3,169

Instructions

a. What was the carrying value of land, buildings, and equipment at the end of 2018?

b. Does Unilever use a conservative or liberal method to depreciate its property, plant, and equipment?

c. What were the actual interest expense and preference dividends paid by the company in 2018?

d. What is Unilever's free cash flow? From the information provided, comment on Unilever's financial flexibility.

Accounting, Analysis, and Principles

Durler Company purchased equipment on January 2, 2021, for $112,000. The equipment had an estimated useful life of 5 years, with an estimated residual value of $12,000. Durler uses straight-line depreciation on all assets. On January 2, 2025, Durler exchanged this equipment, plus $12,000 in cash, for newer equipment. The old equipment has a fair value of $50,000.

Accounting

Prepare the journal entry to record the exchange on the books of Durler Company. Assume that the exchange has commercial substance.

Analysis

How will this exchange affect comparisons of the return on assets ratio for Durler in the year of the exchange compared to prior years?

Principles

How does the concept of commercial substance affect the accounting and analysis of this exchange?

Bridge to the Profession

Authoritative Literature References

1 International Accounting Standard 16, *Property, Plant and Equipment* (London, U.K.: International Accounting Standards Committee Foundation, 2003), par. 6.

2 International Accounting Standard 16, *Property, Plant and Equipment* (London, U.K.: International Accounting Standards Committee Foundation, 2003), par. 7.

3 International Accounting Standard 16, *Property, Plant and Equipment* (London, U.K.: International Accounting Standards Committee Foundation, 2003), par. 16.

4 International Accounting Standard 16, *Property, Plant and Equipment* (London, U.K.: International Accounting Standards Committee Foundation, 2003), par. 22.

5 International Accounting Standard 23, *Borrowing Costs* (London, U.K.: International Accounting Standards Committee Foundation, 2007), paras. 5–6.

6 International Accounting Standard 23, *Borrowing Costs* (London, U.K.: International Accounting Standards Committee Foundation, 2007), paras. 20–25.

7 International Accounting Standard 16, *Property, Plant and Equipment* (London, U.K.: International Accounting Standards Committee Foundation, 2003), paras. 24–26.

8 International Accounting Standard 20, *Accounting for Government Grants and Disclosure of Government Assistance* (London, U.K.: International Accounting Standards Committee Foundation, 2001).

9 International Accounting Standard 37, *Provisions, Contingent Liabilities and Contingent Assets* (London, U.K.: International Accounting Standards Committee Foundation, 2001), paras. 53–58.

Research Case

Your client is in the planning phase for a major plant expansion, which will involve the construction of a new warehouse. The assistant controller does not believe that interest cost can be included in the cost of the warehouse because it is a financing expense. Others on the planning team believe that some interest cost can be included in the cost of the warehouse, but no one could identify the specific authoritative guidance for this issue. Your supervisor asks you to research this issue.

Instructions

Access the IFRS authoritative literature at the IFRS website (you may register for free IFRS access at this site). When you have accessed the documents, you can use the search tool in your Internet browser to respond to the following questions. (Provide paragraph citations.)

a. Is it permissible to capitalize interest (borrowing costs) into the cost of assets? Provide authoritative support for your answer.

b. What are the objectives for capitalizing borrowing costs?

c. Discuss which assets qualify for capitalization.

d. Is there a limit to the amount of interest that may be capitalized in a period?

e. If capitalization of borrowing costs is allowed, what disclosures are required?

Notes

1 Materiality considerations are important in considering items to capitalize. Assume, for example, that Broom Ltd. has spare parts on hand to service any breakdowns in its equipment. Unless the spare parts either separately or in combination are material in amount, expenditures related to spare parts are expensed as incurred even though they provide future benefit.

2 Companies also recognize estimates of dismantling, removing, and site restoration if the company has an obligation that it incurs on acquisition of the asset. For example, **BP** (GBR) indicates that liabilities for decommissioning costs are recognized when the group has an obligation to dismantle or remove a facility or an item of plant and restore the site on which it is located. A corresponding item of property, plant, and equipment of an amount equivalent to the provision is also created. We discuss the recognition and measurement of these assets and related obligations in Chapter 12.

3 Borrowing costs include interest and other costs incurred in connection with borrowing funds. It also includes foreign currency exchange gains and losses related to borrowing in other currencies. **[5]**

4 Also included for borrowing cost capitalization are intangible assets and bearer plants.

5 The valuation approaches that should be used are the market, income, or cost approach, or a combination of these approaches. The *market approach* uses observable prices and other relevant information generated by market transactions involving comparable assets. The *income approach* uses valuation techniques to convert future amounts (for example, cash flows or earnings) to a

single present value amount (discounted). The *cost approach* is based on the amount that currently would be required to replace the service capacity of an asset (often referred to as current replacement cost).

6 Non-monetary assets are items whose price in terms of the monetary unit may change over time. Monetary assets—cash and short- or long-term accounts and notes receivable—are fixed in terms of units of currency by contract or otherwise.

7 The determination of the commercial substance of a transaction requires significant judgment. In determining whether future cash flows change, it is necessary to do one of two things: (1) determine whether the risk, timing, and amount of cash flows arising for the asset received differ from the cash flows associated with the outbound asset; or (2) evaluate whether cash flows are affected with the exchange versus without the exchange. Also, note that if companies cannot determine fair values of the assets exchanged, they should use the carrying value of the asset given up in accounting for the exchange.

8 Recognize that for Jerrod (the dealer), the asset given up in the exchange is considered inventory. As a result, Jerrod records a sale and related cost of goods sold. The used machine received by Jerrod is recorded at fair value.

9 Recognize that there is a distinction between government grants and government assistance. Government assistance can take many forms, such as providing advice related to technical legal or product issues or being a supplier for the company's goods or services. Government grants are a special part of government assistance where financial resources are provided to the company. In rare situations, a company may receive a donation (gift). The accounting for grants and donations is essentially the same. IFRS does provide an option of recording property, plant, and equipment at zero cost although this practice is rarely followed.

10 Using either approach, a question arises about when the amount should be recognized. *IAS 20* says the amount should be recognized when you have "reasonable assurance" that the condition will be satisfied.

11 Both approaches have deficiencies. Reducing the cost of the asset for the grant means that the lab equipment's cost on the statement of financial position may be considered understated. Recording the deferred grant revenue on the credit side of the statement of financial position is problematic as many believe it is not a liability nor is it equity.

12 *IAS 37* permits the recognition of the gain at the point in time when the amount is "virtually certain." If there is a signed agreement between the insurance

company and the company before the cash is received, a receivable could be recognized.